

EASTERN INTERNATIONAL UNIVERSITY
SCHOOL OF COMPUTING AND INFORMATION TECHNOLOGY
DEPARTMENT OF SOFTWARE ENGINEERING



TRƯỜNG ĐẠI HỌC
QUỐC TẾ
MIỀN ĐÔNG
EASTERN
INTERNATIONAL
UNIVERSITY

BACKEND REPORT

FASTFOODHUB

Student(s)

Nguyen Minh Nguyen – 2231209001

Nguyen Minh Man – 2131200038

Nguyen Gia Kiet – 2131200131

Tran Dang Khoa – 2031209002

Huynh Phi Van – 2331200136

Nguyen Hoang Anh Khoa – 2231200100

Ngo Thi Ngoc Huyen – 2331200038

Huynh Thi Huyen Tran – 2331200120

Anh

Supervisor(s)

Nguyen Xuan Cuong

Ho Chi Minh City, June 2025

ABSTRACT

In recent years, the demand for online food ordering services has increased significantly due to the rapid development of information technology and changing consumer lifestyles. Customers increasingly expect convenient and efficient ways to browse menus, place orders, and manage their purchases without relying on traditional ordering methods. However, manual ordering processes may lead to communication errors, inefficient order management, and difficulties in maintaining accurate business information.

This project presents FastFoodHub, a web-based fast food ordering system designed to simplify the food ordering process and improve restaurant management. The system allows customers to browse food categories, search for products, add items to a shopping cart, place orders, and review their order history. In addition, the application provides an administrative module that enables administrators to manage food categories, products, customer accounts, and order information through a centralized platform.

FastFoodHub is developed using a structured software development process that includes requirement analysis, system design, database modeling, implementation, and testing. The system is implemented using ASP.NET Web Forms, C#, MySQL, Bootstrap, and JavaScript to provide a responsive and user-friendly interface.

The primary objective of FastFoodHub is to provide customers with convenient and reliable online ordering experience while helping restaurants improve operational efficiency and simplify management activities. By integrating customer-oriented and administrative functionalities into a single platform, the system aims to contribute to the digital transformation of restaurant businesses and enhance overall service quality.

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to all those who have contributed to the successful completion of our project, FastFoodHub – a web-based fast food ordering system. Throughout the development of this project, we received valuable guidance, encouragement, and support from many individuals and organizations who helped transform our ideas into a practical and functional application.

First and foremost, we would like to extend our deepest appreciation to Mr. Nguyen Xuan Cuong, our supervisor, for his invaluable guidance and mentorship throughout the project. His expertise, constructive feedback, and continuous support played a significant role in guiding us through every stage of the development process. His encouragement and advice helped us overcome technical challenges and improve both the quality and usability of the system.

We would also like to express our sincere thanks to the Department of Software Engineering at Eastern International University for providing us with the knowledge, resources, and academic environment necessary for completing this project. The dedication and support of our lecturers and academic staff have greatly contributed to the development of our technical knowledge, professional skills, and problem-solving abilities.

In addition, we would like to thank all individuals who participated in the testing and evaluation of the system. Their feedback and suggestions were valuable in helping us identify areas for improvement and enhance the overall performance and user experience of the application.

Finally, we would like to express our heartfelt gratitude to our families and friends for their unwavering encouragement, patience, and support throughout the development of this project. Their motivation and understanding gave us the strength and determination needed to successfully complete this work.

We are truly grateful to everyone who contributed, directly or indirectly, to the completion of this project.

TABLE OF CONTENTS

ABSTRACT	i
ACKNOWLEDGEMENT	ii
TABLE OF CONTENTS	iii
LIST OF FIGURES	vi
LIST OF TABLES	viii
LIST OF ABBREVIATIONS	ix
CHAPTER 1. INTRODUCTION	1
1.1. Motivation	1
1.2. Objective	1
1.3. Project Overview	2
CHAPTER 2. RELATED WORKS	3
2.1. Application Market Survey	3
2.2. Target Users and Scope	3
2.3. Core Features	3
2.4. Integration and Technology	4
2.5. User Experience	4
2.6. Benefits	4
2.7. Key Differentiators	5
2.8. Frontend Technologies	5
2.8.1. HTML	5
2.8.2. CSS	5
2.8.3. Bootstrap	5
2.8.4. JavaScript	6
2.8.5. jQuery DataTables	6
2.8.6. Backend Technologies	6
2.8.7. ASP.NET Web Forms	6
2.8.8. C#	6
2.8.9. Visual Studio	6
2.8.10. Database Technologies	6
CHAPTER 3. System Design	8
3.1. User Case	8
3.1.1. General user diagram	8
3.1.2. Admin Module Use Case	9

3.1.3. Guest Module Use Case	10
3.1.4. Customer Module Use Case	11
3.2. Activiti Diagram	12
3.2.1. Payment Transaction Process	12
3.2.2. Add to Cart & Checkout Process	13
3.2.3. Admin Update Order Status Process	14
3.2.4. User Authentication Process	15
3.2.5. Product Management Process	16
3.2.6. Selling Report Generation Process	17
3.3. System Workflow	18
3.3.1. Baseflow Guest Workflow	18
3.3.2. Baseflow Admin Workflow	19
3.3.3. Baseflow Customer Workflow	20
3.4. Database	21
3.4.1. User's Table	21
3.4.2. Categories Table	22
3.4.3. Products Table	22
3.4.4. Carts Table	22
3.4.5. Orders Table	23
3.4.6. Payment Table	23
3.4.7. Contact Table	23
3.4.8. Relationships Between Entities	24
CHAPTER 4. RESULT	26
4.1. Login and Sign-up Screen	26
4.1.1. Login Screen	26
4.1.2. Register Screen	27
4.2. Main Page	28
4.2.1. Home Page	28
4.2.2. Food Menu	30
4.2.3. About Us Page	31
4.2.4. Contact Page	32
4.2.5. User Profile Page	33
4.2.6. Cart Page	35
4.2.7. Payment Page	36
4.2.8. Order Success Page	37

4.3. Admin Page	40
4.3.1. Admin Dashboard	40
The dashboard is designed with a clean and user-friendly layout that enhances usability and supports effective decision-making. By centralizing both analytical information and management functions in a single interface, the Admin Dashboard improves operational efficiency, simplifies administrative tasks, and contributes to the overall success of the online food ordering system.	41
4.3.2. Categories Management	41
4.3.3. Product Management	42
4.3.4. Order Status Management	43
4.3.5. User Management	44
4.3.6. Contact Management	45
4.3.7. Selling Report	46
CHAPTER 5. DISCUSSION	47
5.1. Project Outcomes	47
5.2. Performance Evaluation	47
5.3. Discussion	48
5.3.1. Customer Features	48
5.3.2. Administrative Management	48
5.3.3. Sales Reporting	48
CHAPTER 6. COMCLUSION AND FUTURE IMPROVEMENT	50
REFERENCES	51

LIST OF FIGURES

Figure 1 Foodie Ordering System Overview	8
Figure 2 Admin Module Use Case Diagram	9
Figure 3 Guest Module Use Case Diagram	10
Figure 4 Customer Module Use Case Diagram	11
Figure 5 Payment Transaction Process	12
Figure 6 Add Cart & Checkout Process	13
Figure 7 Admin Update Order Status Process	14
Figure 8 User Authentication Process	15
Figure 9 Product Management Process	16
Figure 10 Selling Report on Generation Process	17
Figure 11 Activity Diagram for Guest User Interaction with the System	18
Figure 12 Activity Diagram for Admin Interaction with the System	19
Figure 13 Activity Diagram for Customer Interaction with the System	20
Figure 14 Relationship Diagram	21
Figure 15 Login Page	26
Figure 16 Register Page	27
Figure 17 Home Page	28
Figure 18 Home Page	29
Figure 19 Food Menu Page	30
Figure 20 About Us Page	31
Figure 21 Contact Form Page	32
Figure 22 User Profile Page	33
Figure 23 Purchase History Page	34
Figure 24 Cart Page	35
Figure 25 Payment (Credit Card Method)	36
Figure 26 Payment (Cash of Delivery Method)	37
Figure 27 Order Information Page	37
Figure 28 Print the Order Mode	38
Figure 29 Watch PDF Order Invoice	39
Figure 30 Admin Dashboard	40
Figure 31 Categories Management	41
Figure 32 Product Management	42
Figure 33 Order's Update Status	43

Figure 34 Users Management	44
Figure 35 Contact Management	45
Figure 36 Selling Reports	46

LIST OF TABLES

Table 1 User's Table	21
Table 2 Categories Table	22
Table 3 Product Table	22
Table 4 Cart Table	22
Table 5 Order Table	23
Table 6 Payment Table	23
Table 7 Contact Table	23
Table 8 Relationships Between Entities Table	24

LIST OF ABBREVIATIONS

No.	Term	Meaning
1	ASP.NET	Active Server Pages .NET
2	C#	C Sharp Programming Language
3	DBMS	Database Management System
4	IDE	Integrated Development Environment
5	UI	User Interface
6	UX	User Experience
7	SQL	Structured Query Language
8	MySQL	My Structured Query Language
9	HTML	Hypertext Markup Language
10	CSS	Cascading Style Sheets
11	JS	JavaScript
12	CRUD	Create, Read, Update, Delete
13	API	Application Programming Interface
14	ERD	Entity Relationship Diagram
15	UML	Unified Modeling Language
16	COD	Cash On Delivery
17	PK	Primary Key
18	FK	Foreign Key
19	INT	Integer
20	VARCHAR	Variable Character

CHAPTER 1. INTRODUCTION

1.1. Motivation

The motivation behind developing FastFoodHub stems from the need to address the inefficiencies and challenges faced by both customers and restaurant administrators in managing food ordering services and daily operations. In today's fast-paced lifestyle, the demand for convenient and efficient food ordering solutions is continuously increasing. However, traditional methods of ordering food, such as in-store purchases and phone calls, often result in long waiting times, communication errors, and inefficient order management.

Customers frequently experience difficulties when searching for menus, placing orders, and tracking their order status. The lack of a centralized platform for browsing food items and making payments adds inconvenience to their overall experience. On the other hand, restaurant administrators struggle with managing menus, processing customer orders, monitoring sales, and handling operational tasks efficiently. The absence of streamlined processes and effective communication channels further complicates restaurant management and reduces service quality.

1.2. Objective

In this context, FastFoodHub was conceived to leverage advancements in web technologies to improve the way food ordering services are provided and managed. The primary objective is to develop an application that not only digitizes traditional ordering processes but also provides a comprehensive set of tools to enhance customer experience and streamline restaurant operations.

For customers, FastFoodHub offers the convenience of browsing food menus, searching for products, adding items to the shopping cart, placing orders, and viewing order history. It allows customers to complete transactions efficiently while reducing the time and effort required in traditional ordering methods.

For restaurant administrators, FastFoodHub provides functionalities to manage food categories, update menu items, process customer orders, and monitor business activities. By offering a centralized platform for management and communication, FastFoodHub aims to improve operational efficiency, reduce manual workloads, and enhance service quality.

The ultimate goal of FastFoodHub is to create a convenient and integrated food ordering experience that benefits both customers and restaurant businesses. By addressing existing challenges and leveraging modern technologies, FastFoodHub aspires to improve service efficiency, enhance customer satisfaction, and contribute to the digital transformation of the food service industry.

1.3. Project Overview

FastFoodHub is a comprehensive web-based fast food ordering application designed to improve customer experience and simplify restaurant management. It integrates key features such as menu browsing, product searching, shopping cart management, order processing, and administrative functions into a unified platform. By digitizing traditional food ordering procedures, FastFoodHub aims to increase operational efficiency and reduce the limitations associated with manual order processing.

The application consists of two primary user interfaces: one for customers and one for administrators. For customers, FastFoodHub provides an intuitive interface to browse food menus, view product details, place orders, and access order history at any time. The platform also enables customers to manage their accounts and complete transactions conveniently.

For administrators, FastFoodHub offers a dedicated management interface that allows efficient control of restaurant operations. Administrators can manage food items, categories, customer accounts, and order information through a centralized dashboard. The system also provides functionalities for updating product information, monitoring order status, and maintaining business data, thereby improving management efficiency and reducing administrative workloads.

FastFoodHub leverages modern web technologies and database management systems to ensure that information is accurate, secure, and easily accessible. With a focus on user-friendly design and robust functionality, the application aims to provide a streamlined, efficient, and scalable solution to address the evolving needs of restaurant businesses and their customers.

Unlike large-scale food delivery platforms, FastFoodHub primarily focuses on online food ordering and restaurant management. Ultimately, the system is intended to provide a convenient, accessible, and customer-oriented ordering experience while helping restaurants improve operational efficiency through a centralized administrative platform.

CHAPTER 2. RELATED WORKS

2.1. Application Market Survey

– McDonNowadays, several applications provide functionalities like those offered by the proposed system. Some representative examples are as follows:

- + McDonald's Online Ordering System.
- + KFC Online Ordering System.
- + Domino's Pizza Online Ordering System.
- + Pizza Hut Online Ordering System
- + Burger King Online Ordering System
- + Subway Online Ordering System

– Analysis and Comparison of McDonald's Online Ordering System and FastFoodHub.

Among the applications mentioned above, one of the most prominent systems is the McDonald's Online Ordering System. Its popularity and extensive range of features make it a suitable reference system for comparison with the proposed FastFoodHub application.

2.2. Target Users and Scope

– McDonald's Online Ordering System:

The McDonald's Online Ordering System is designed to serve customers who wish to browse menus and place orders conveniently through a digital platform. It also supports restaurant administrators in managing products and customer orders efficiently.

– FastFoodHub:

FastFoodHub focuses on providing a web-based food ordering system for customers and restaurant administrators. The system allows customers to browse menus, place orders, and manage their order history, while administrators can manage food categories, products, and customer orders.

2.3. Core Features

– McDonald's Online Ordering System:

The system offers menu browsing, food searching, shopping cart management, online ordering, and payment functionalities. Customers can customize their orders and view detailed product information.

– FastFoodHub:

FastFoodHub provides essential features such as food menu browsing, product searching, shopping cart management, and online order placement. Customers can view detailed information about food items and manage their order history. In addition, the system includes

an administrative module that allows administrators to manage food categories, products, customer accounts, and order information efficiently. By focusing on streamlined ordering workflows and user-friendly interfaces, FastFoodHub aims to improve customer satisfaction and enhance restaurant management efficiency.

2.4. Integration and Technology

– McDonald's Online Ordering System:

The system employs modern web technologies and centralized databases to provide stable and efficient food ordering services.

– FastFoodHub:

FastFoodHub employs a web-based architecture developed using ASP.NET Web Forms, C#, and MySQL. The system focuses on usability and maintainability to support food ordering and restaurant management.

2.5. User Experience

– McDonald's Online Ordering System:

The McDonald's Online Ordering System provides a user-friendly and responsive interface designed to ensure accessibility for customers. Its intuitive layout and straightforward ordering process allow users to browse menus, customize their meals, and place orders conveniently. The system is suitable for restaurants that aim to provide a seamless and efficient digital ordering experience.

– FastFoodHub:

FastFoodHub emphasizes simplicity and ease of use through dedicated interfaces for both customers and administrators. Customers can conveniently browse menus and place orders, while administrators are provided with a management dashboard for handling products, categories, customer accounts, and order information. The design is particularly beneficial for small and medium-sized restaurants aiming to improve operational efficiency and customer satisfaction.

2.6. Benefits

– McDonald's Online Ordering System:

The McDonald's Online Ordering System simplifies the ordering process and enhances customer experience through online ordering and payment functionalities. It also helps restaurants improve service quality and increase operational efficiency by reducing manual order processing.

– FastFoodHub:

FastFoodHub focuses on restaurant-centered operations and online ordering workflows. It is an ideal solution for small and medium-sized restaurants that require a simple yet effective platform to manage food items, customer orders, and business operations. In addition to customer functionalities, the system provides an administrative module with centralized management capabilities, allowing administrators to efficiently manage products, categories, users, and orders.

2.7. Key Differentiators

– McDonald's Online Ordering System:

The McDonald's Online Ordering System is a mature and large-scale platform with a well-established ordering infrastructure. Its strengths lie in providing a smooth ordering experience and supporting efficient restaurant operations.

– FastFoodHub:

FastFoodHub focuses on restaurant-centered operations and online ordering workflows. It is an ideal solution for small and medium-sized restaurants that require a simple yet effective platform to manage food items, customer orders, and business operations. Unlike large commercial platforms, FastFoodHub emphasizes usability,

2.8. Frontend Technologies

FastFoodHub is a web-based fast food ordering application, and its frontend technologies were carefully selected to provide a responsive, maintainable, and user-friendly interface. The chosen technologies include HTML, CSS, Bootstrap, JavaScript, and jQuery DataTables. Each technology plays a specific role in the development process.

2.8.1. HTML

HTML provides the structural foundation for the FastFoodHub application. Every page, including the home page, product pages, shopping cart, and administrative pages, is ultimately rendered as HTML in the browser. By using semantic HTML elements, the system ensures better accessibility, browser compatibility, and maintainability.

2.8.2. CSS

CSS is used to control the appearance and layout of the application. It helps provide a consistent visual design and improves the user's experience. Custom styles are applied to enhance the presentation of menus, tables, forms, and navigation components throughout the system.

2.8.3. Bootstrap

Bootstrap is a popular front-end framework that provides responsive design capabilities and pre-built user interface components. It enables the system to adapt to different screen sizes

and devices while ensuring a consistent appearance. Components such as navigation bars, forms, buttons, and tables are widely used in the FastFoodHub system.

2.8.4. JavaScript

JavaScript is used to provide interactive functionalities and improve user experience. It handles dynamic content updates, client-side validation, and interactions between users and the web pages, making the application more responsive and efficient.

2.8.5. jQuery DataTables

jQuery DataTables is used to enhance table functionalities within the administrative interface. It provides features such as searching, sorting, pagination, and data export, allowing administrators to manage products, categories, and orders more efficiently.

2.8.6. Backend Technologies

FastFoodHub requires a stable backend to process customer requests and manage restaurant operations. The backend technologies used in this project include ASP.NET Web Forms, C#, and Visual Studio.

2.8.7. ASP.NET Web Forms

ASP.NET Web Forms is a web application framework developed by Microsoft. It enables rapid development through server controls and event-driven programming. In this project, ASP.NET Web Forms is used to implement the business logic and manage interactions between users and the system.

2.8.8. C#

C# is an object-oriented programming language developed by Microsoft. It is used to implement functionalities such as user authentication, product management, shopping cart operations, and order processing. Its reliability and extensive libraries make it suitable for developing web applications.

2.8.9. Visual Studio

Visual Studio is an integrated development environment (IDE) used for developing the FastFoodHub system. It provides powerful tools for coding, debugging, testing, and maintaining applications. Visual Studio improves productivity and simplifies the software development process.

2.8.10. Database Technologies

MySQL is a widely used relational database management system. In FastFoodHub, MySQL serves as the backend database, storing structured data such as user accounts, categories, products, shopping carts, orders, and payment information.

MySQL was chosen because of its reliability, ease of use, and compatibility with ASP.NET applications. It supports efficient data storage and retrieval while ensuring data consistency through relationships and constraints. In addition, its open-source nature and extensive community support make MySQL a practical and cost-effective solution for developing the FastFoodHub system.

CHAPTER 3. System Design

3.1. User Case

3.1.1. General user diagram

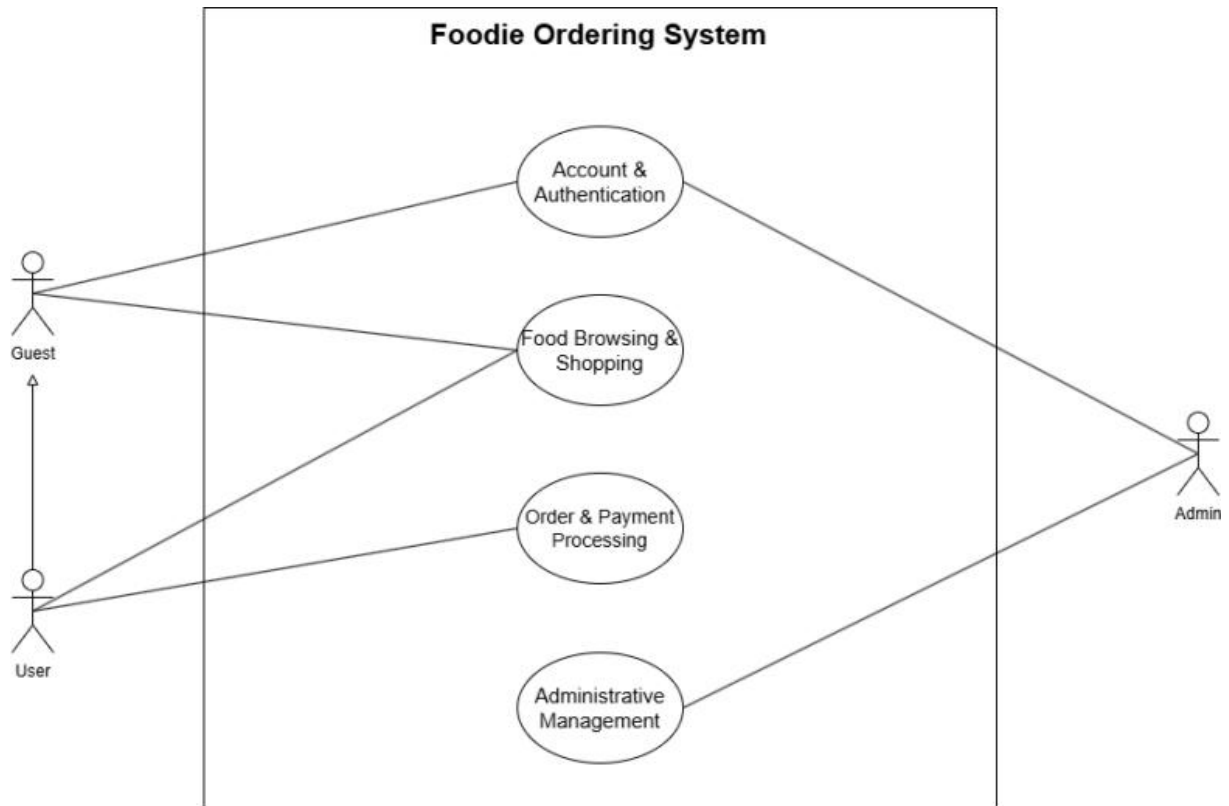


Figure 1 Foodie Ordering System Overview

- **Actors:** Guest, User, Admin.
- **Description:** This diagram provides a high-level architectural view of the system, defining the primary functional modules and the interaction boundaries between different actor types.
- **Flow:**
 - + The guest and User interact with the system for Authentication, Food Browsing, Shopping, and Order/Payment Processing.
 - + Admin interacts with the system primarily through Administrative Management.
 - + The system acts as the container for these four core functional areas.
- **Goal:** To define the main functional modules of the Foodie Ordering System and establish the scope of user interactions for each actor.

3.1.2. Admin Module Use Case

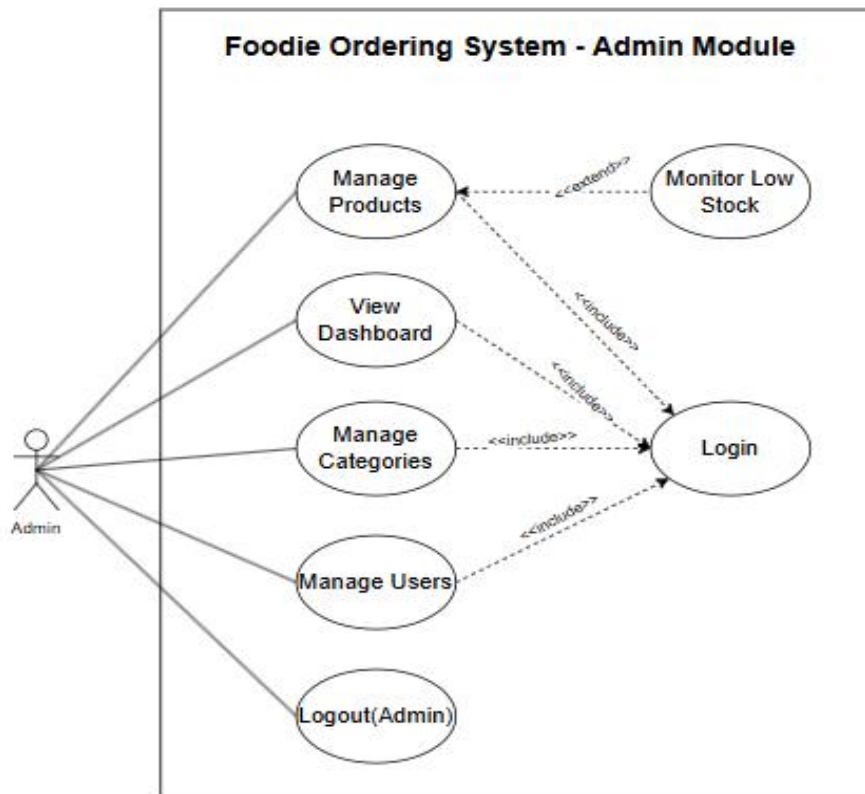


Figure 2 Admin Module Use Case Diagram

- **Actors:** Admin.
- **Description:** This module delineates the administrative functionalities required to operate and maintain the Foodie Ordering System, focusing on operational oversight, product inventory control, and user management.
- **Flow:**
 - + The admin must authenticate via the "Login" use case to access protected system features.
 - + Upon successful authentication, the admin can perform essential management tasks, including managing products (with an extension feature to monitor low stock), managing product categories, and managing user accounts.
 - + The admin can access the system dashboard ("View Dashboard") to monitor activities.
 - + The admin can terminate the session securely using the "Logout" functionality.
- **Goal:** To provide the administrator with centralized control over inventory, system analytics, and user access rights to ensure efficient ordering system management.

3.1.3. Guest Module Use Case

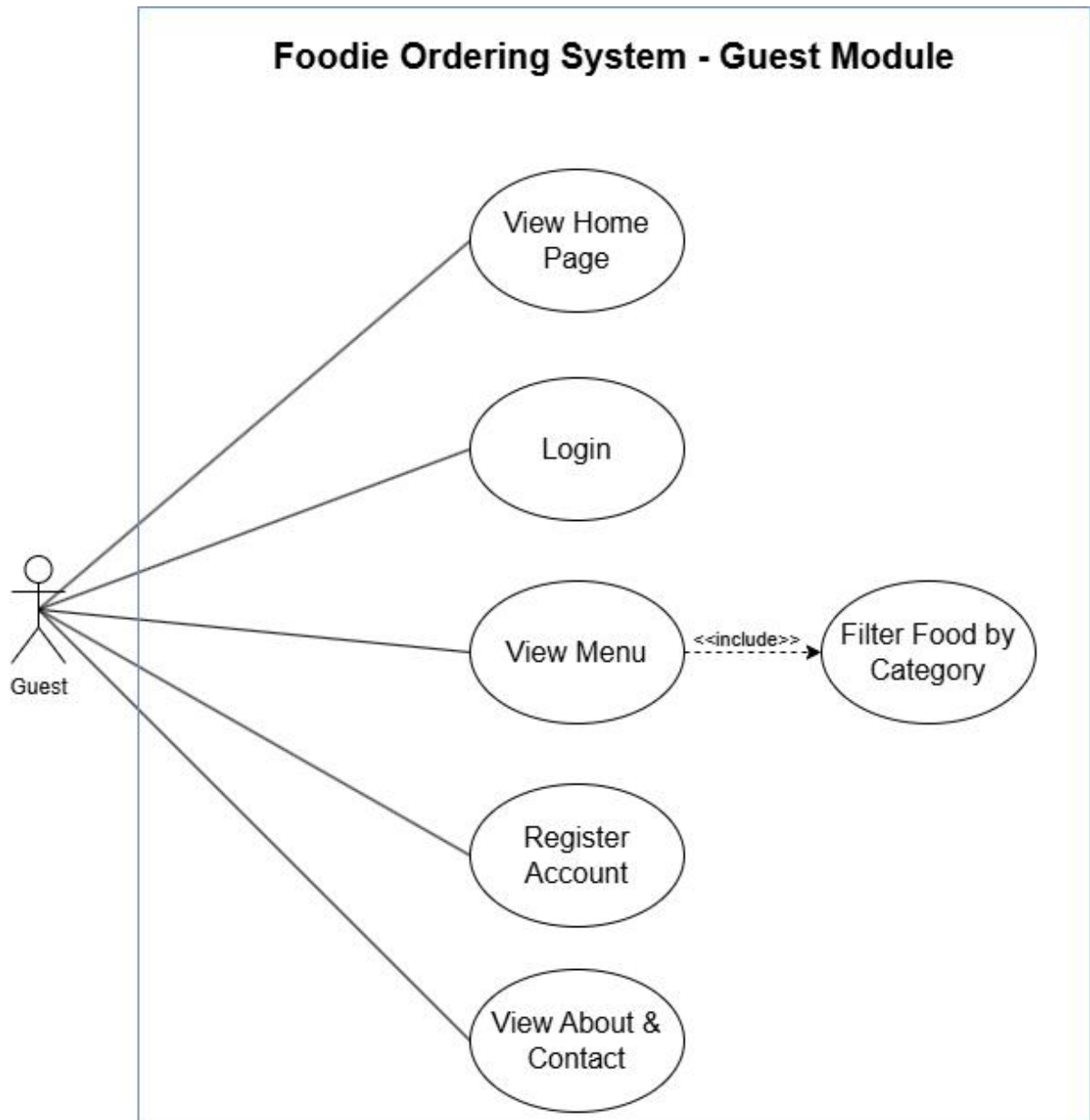


Figure 3 Guest Module Use Case Diagram

- **Actors:** Guest.
- **Description:** This module covers the public-facing functionalities available to users who have not yet logged in or registered an account.
- **Flow:**
 - + The guest accesses the home page and views the menu.
 - + The guest interacts with the menu by filtering food items by category.
 - + The guest views contact and "About" information to understand the system.
 - + When ready, the guest chooses to register a new account or log in to access protected features.

– **Goal:** To allow public access to system information and provide entry points for user registration and authentication.

3.1.4. Customer Module Use Case

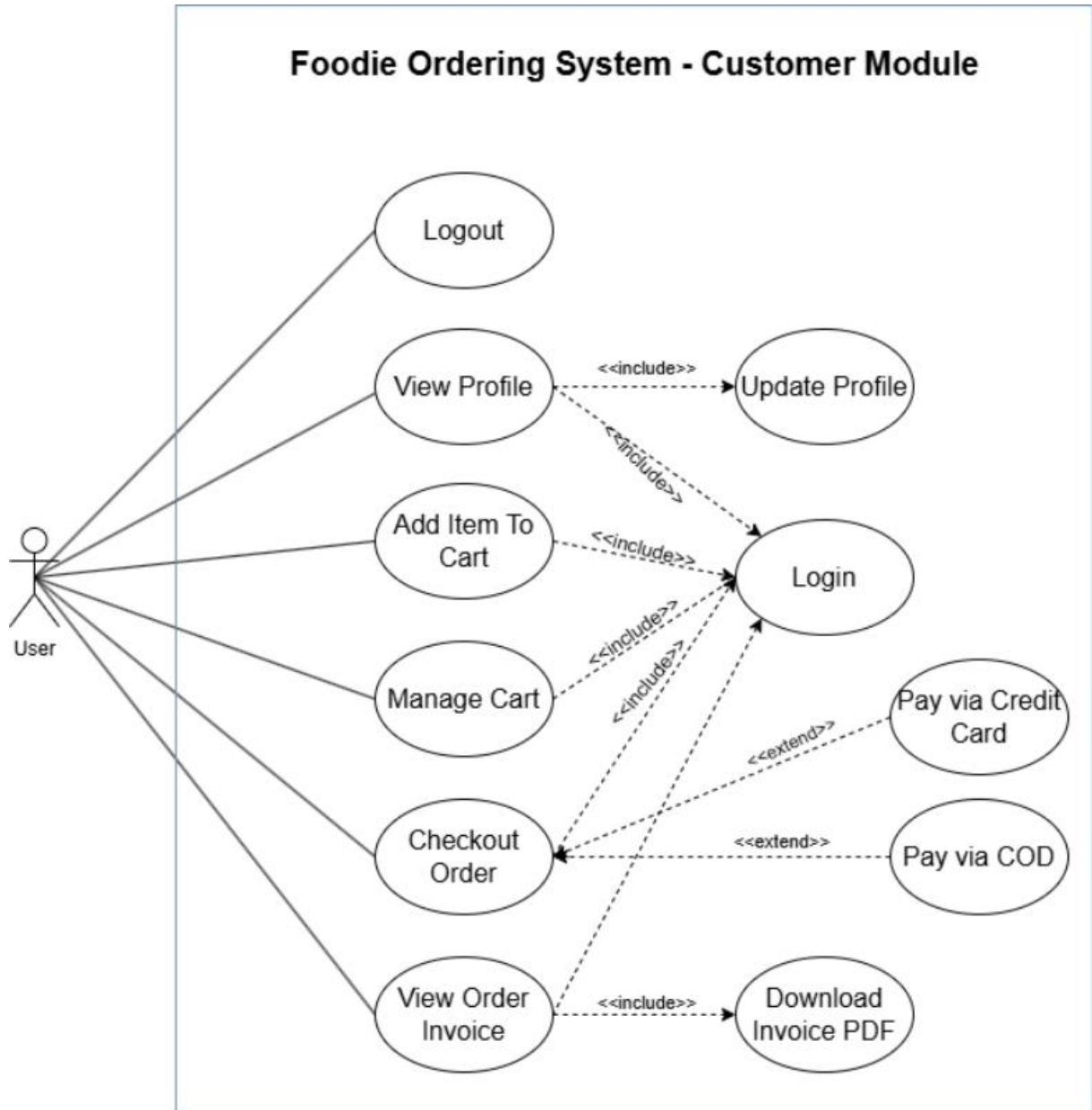


Figure 4 Customer Module Use Case Diagram

– **Actors:** User.

– **Description:** This module encompasses the functionalities available to authenticated users, focusing on personal profile management, shopping cart operations, and order processing.

– **Flow:**

+ The user accesses the system and logs in (required for cart and profile actions).

- + The user manages their shopping cart (Add Item, View Cart) and updates their profile information.
- + The user proceeds to check out, selects a payment method (Credit Card or COD), and completes the order.
- + The system generates an invoice, which the user can then view or download.
 - **Goal:** To provide a personalized environment for users to manage their accounts, purchase food items, and track order history.

3.2. Activiti Diagram

3.2.1. Payment Transaction Process

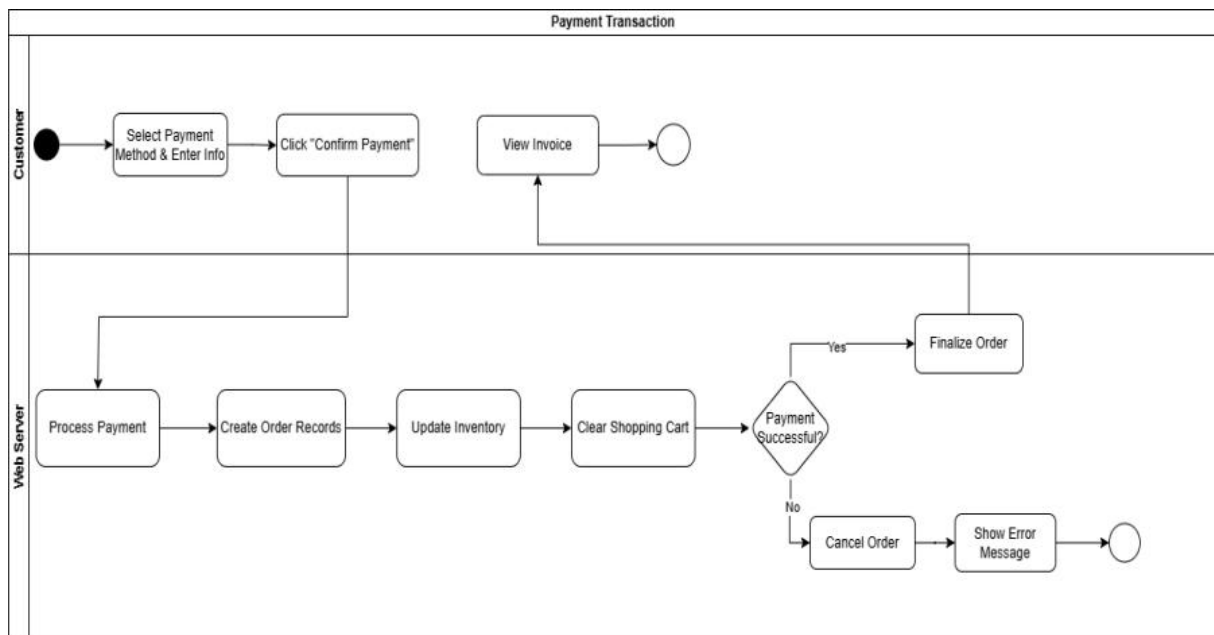


Figure 5 Payment Transaction Process

- **Participants:** Customer, Web Server.
- **Process Flow:**
 - + **Start:** The process begins when the customer navigates to the checkout phase.
 - + **Select Payment Method & Enter Info:** The customer inputs their delivery address and selects a payment method (Credit Card or Cash on Delivery).
 - + **Click "Confirm Payment":** The customer submits the order request.
 - + **Process Payment:** The system opens a database transaction and executes the save payment procedure to store payment details.
 - + **Create Order Records:** The system retrieves the cart items and loops through them to generate order records in the database.
 - + **Update Inventory:** Within the same loop, the system deducts the purchased quantities from the products table.

- + **Clear Shopping Cart:** The system deletes the processed items from the user's cart.
- + **Payment Successful:** The system verifies if all database operations in the transaction were executed without any exceptions.
- + **Cancel Order & Show Error Message:** If an error occurs, the system triggers a Rollback () to cancel all database changes and displays an error alert to the customer.
- + **Finalize Order:** If no errors occur, the system triggers a Commit () to permanently save the data.
- + **View Invoice:** The customer is redirected to the invoice page where they can view or download their order receipt.
- + **End:** The transaction is completed.

3.2.2. Add to Cart & Checkout Process

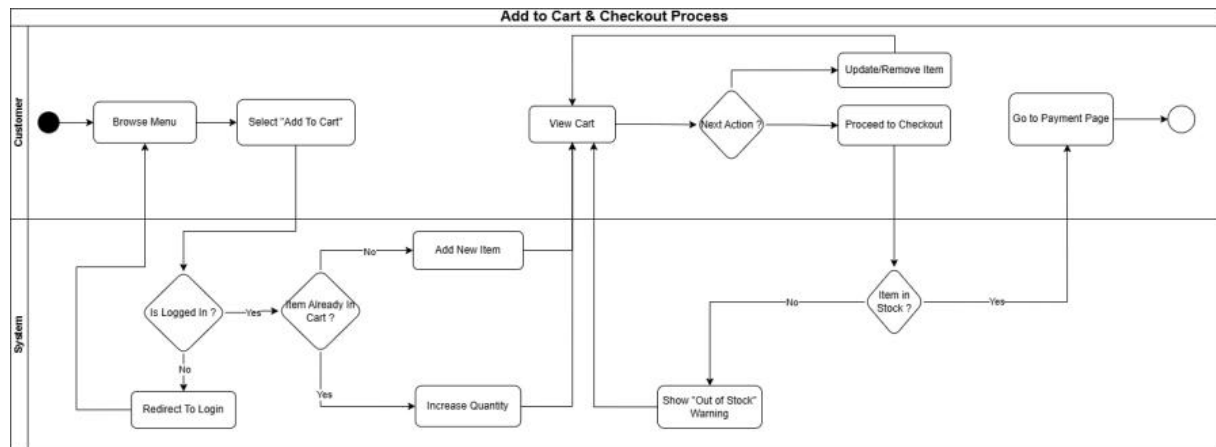


Figure 6 Add Cart & Checkout Process

– **Participants:** Participants: Customer, Web Server.

– **Process Flow:**

- + **Start:** The process begins when the customer accesses the system.
- + **Browse Menu:** The customer views the available food items on the menu page.
- + **Select "Add to Cart":** The customer clicks the button to add a specific item to their order.
- + **Is Logged In:** The system verifies if the user has an active login session.
- + **Redirect To Login:** If the session is invalid or null, the system redirects the customer to the login page.
- + **Item Already in Cart:** The system queries the carts table in the database to check if the selected product already exists for this user.
- + **Add New Item:** If the item is not found, the system executes an INSERT command to add the product with a quantity of 1.

- + **Increase Quantity:** If the item already exists, the system executes an UPDATE command to increase the quantity.
- + **View Cart:** The customer navigates to the shopping cart interface.
- + **Next Action:** The customer decides whether to modify their current cart or proceed to check out.
- + **Update/Remove Item:** If the customer changes quantities or removes items, the system updates the database accordingly and refreshes the cart view.
- + **Proceed to Checkout:** The customer clicks the checkout button.
- + **Item in Stock:** The system validates if the requested cart quantity exceeds the available product quantity ($\text{productQuantity} \geq \text{cartQuantity}$).
- + **Show "Out of Stock" Warning:** If stock is insufficient, the system blocks the action and displays a warning message to the customer.
- + **Go to Payment Page:** If the stock validation passes, the system successfully redirects the customer to the payment page.
- + **End:** The process concludes as it transitions to the Payment Transaction flow.

3.2.3. Admin Update Order Status Process

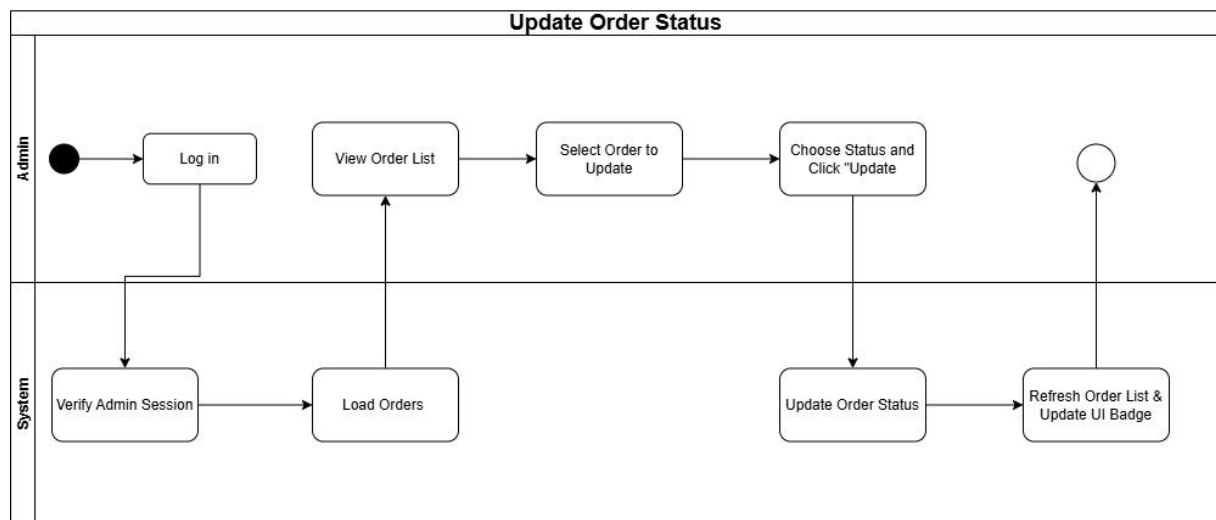


Figure 7 Admin Update Order Status Process

- **Participants:** Admin, System.
- **Process Flow:**
- + **Start:** The process begins when the admin navigates to the Order Status page.
- + **Verify Admin Session:** The system checks if the current user has the correct administrative privileges.
- + **Load Orders:** The system executes the `sp_GetOrderHistory` procedure to retrieve pending and completed orders from the database.

- + **View Order List:** The admin views the retrieved data on the dashboard grid.
- + **Select Order & Choose Status:** The Admin selects a specific order and chooses a new fulfillment status (e.g., Dispatched or Delivered).
- + **Click "Update":** The Admin submits the status change.
- + **Execute Update:** The system calls the sp_UpdateOrderStatus stored procedure to permanently update the order's status in the database.
- + **Refresh UI:** The system reloads the order list and updates the status badge color to reflect the recent changes.
- + **End:** The fulfillment process is successfully completed.

3.2.4. User Authentication Process

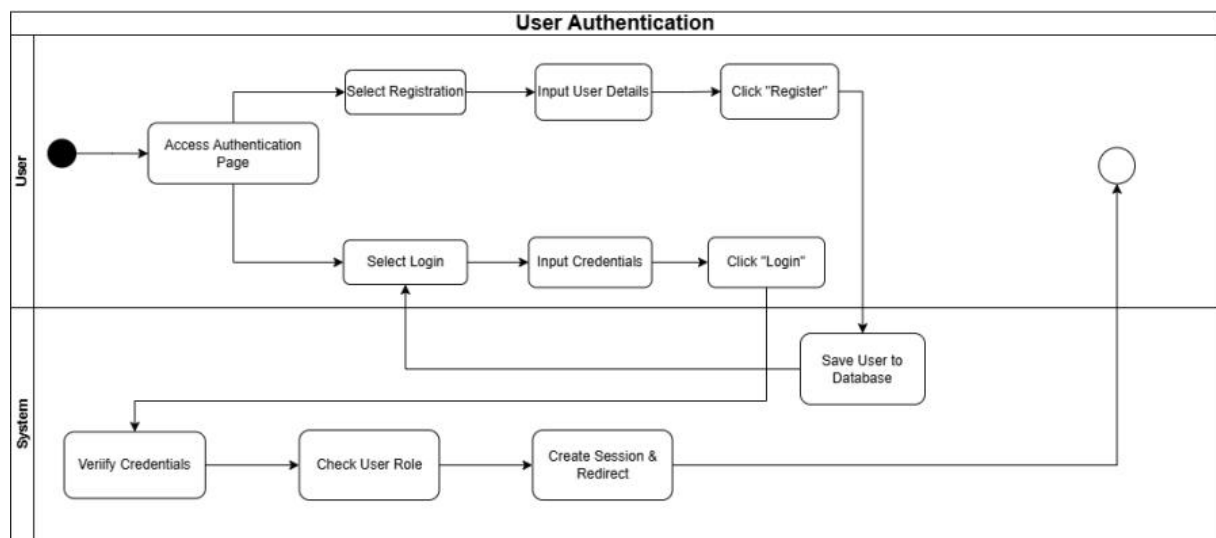


Figure 8 User Authentication Process

- **Participants:** User, System.
- **Process Flow:**
 - + **Start:** The process begins when the user accesses the authentication page.
 - + **Select Registration:** If the user is new, they select the registration option, fill in their personal details, and submit the form. The system saves the new record to the database and redirects them to login.
 - + **Input Credentials:** The user inputs their username and password and clicks "Login".
 - + **Verify Credentials:** The system queries the database to validate the credentials provided.
 - + **Check User Role:** Upon successful verification, the system checks if the account belongs to an Admin or a Customer.
 - + **Create Session & Redirect:** The system initializes the user session (Session["userId"] or Session["admin"]) and routes the user to the appropriate landing page (Dashboard for Admin, Home for Customer).

+ **End:** The authentication process is completed.

3.2.5. Product Management Process

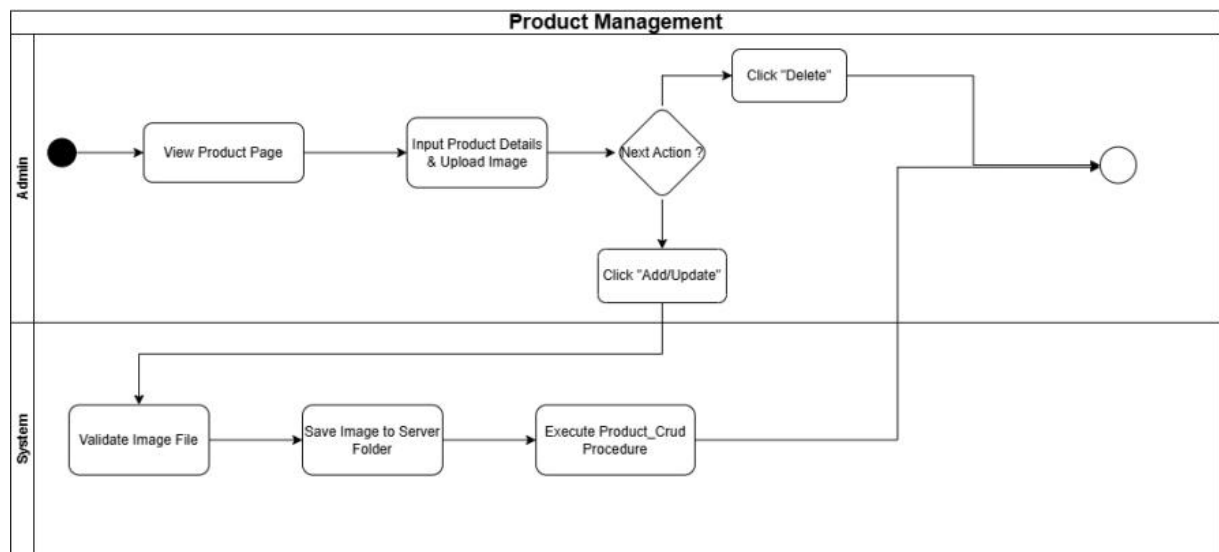


Figure 9 Product Management Process

– **Participants:** Admin, System.

– **Process Flow:**

- + **Start:** The admin navigates to the Product Management module.
- + **Input Details & Image:** The Admin fills out the product information (name, price, category, quantity) and uploads a product image.
- + **Action - Add/Update:** If adding or updating, the system validates the image extension, saves the physical image file to the server folder, and executes the Product_Crud stored procedure to update the database.
- + **Action - Delete:** If deleted, the system removes the specific product record from the database.
- + **Refresh Grid:** The system clears the input form, reloads the updated product list, and displays a success notification.
- + **End:** The product management task is finalized.

3.2.6. Selling Report Generation Process

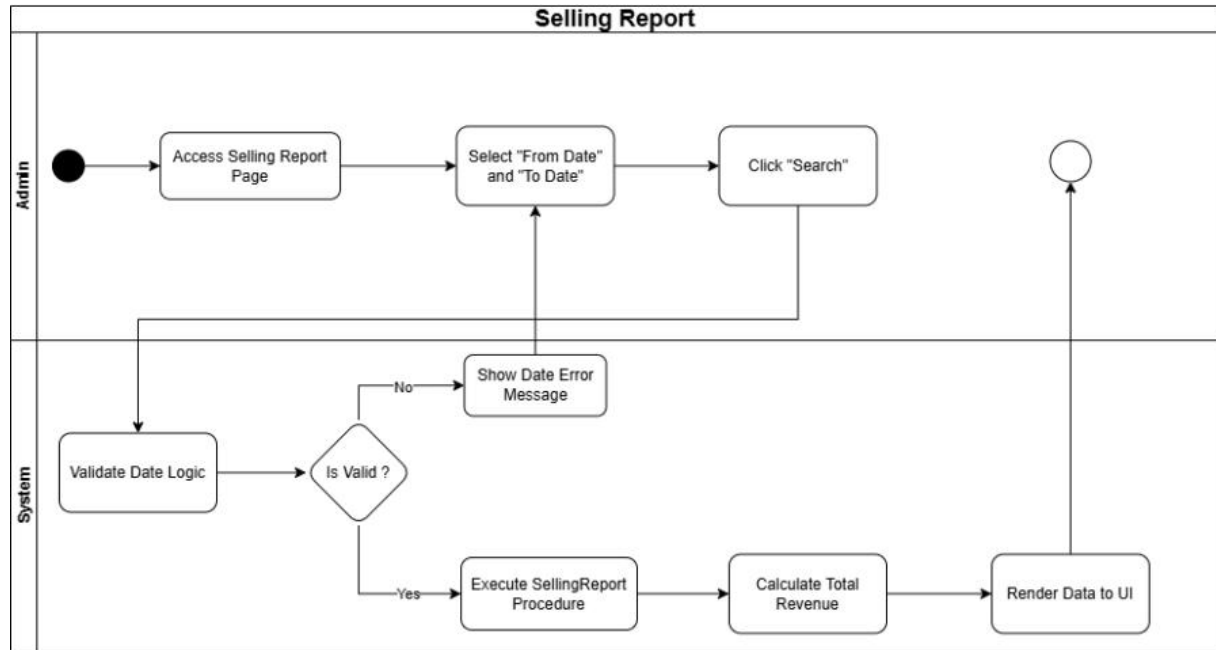


Figure 10 Selling Report on Generation Process

– **Participants:** Admin, System.

– **Process Flow:**

- + **Start:** The admin accesses the Selling Report dashboard.
- + **Select Dates:** The admin inputs the desired timeframe by selecting a "From Date" and a "To Date", then clicks the Search button.
- + **Validate Date Logic:** The system validates the inputs to ensure "To Date" is not in the future and "From Date" is not greater than "To Date".
- + **Show Error:** If validation fails, the system halts the operation and alerts the Admin to correct the dates.
- + **Execute Query:** If valid, the system calls the Selling Report stored procedure to retrieve completed orders within the timeframe.
- + **Calculated Revenue:** The system iterates through the fetched records to calculate the grand total revenue.
- + **Render Data:** The system populates the data grid and displays the total amount.
- + **End:** The report generation is completed.

3.3. System Workflow

3.3.1. Baseflow Guest Workflow

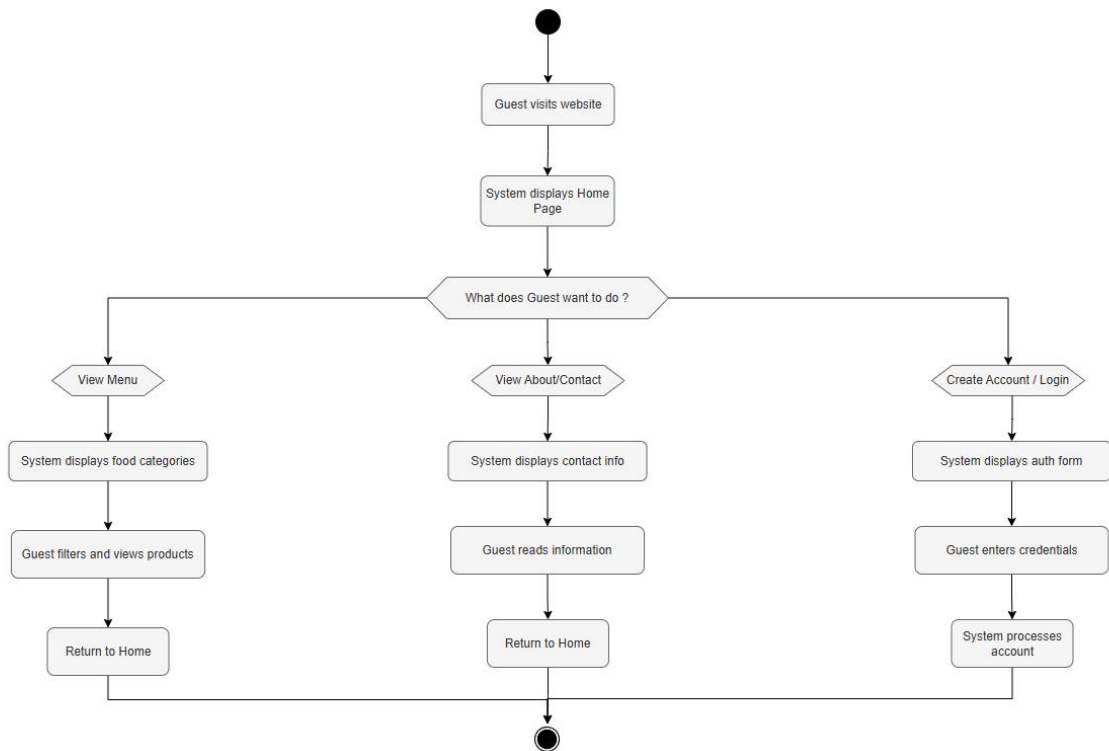


Figure 11 Activity Diagram for Guest User Interaction with the System

The diagram shows how a guest visits the website, browses the available menu and food categories, views system information, and can register or log in to an account if they want to access shopping features.

3.3.2. Baseflow Admin Workflow

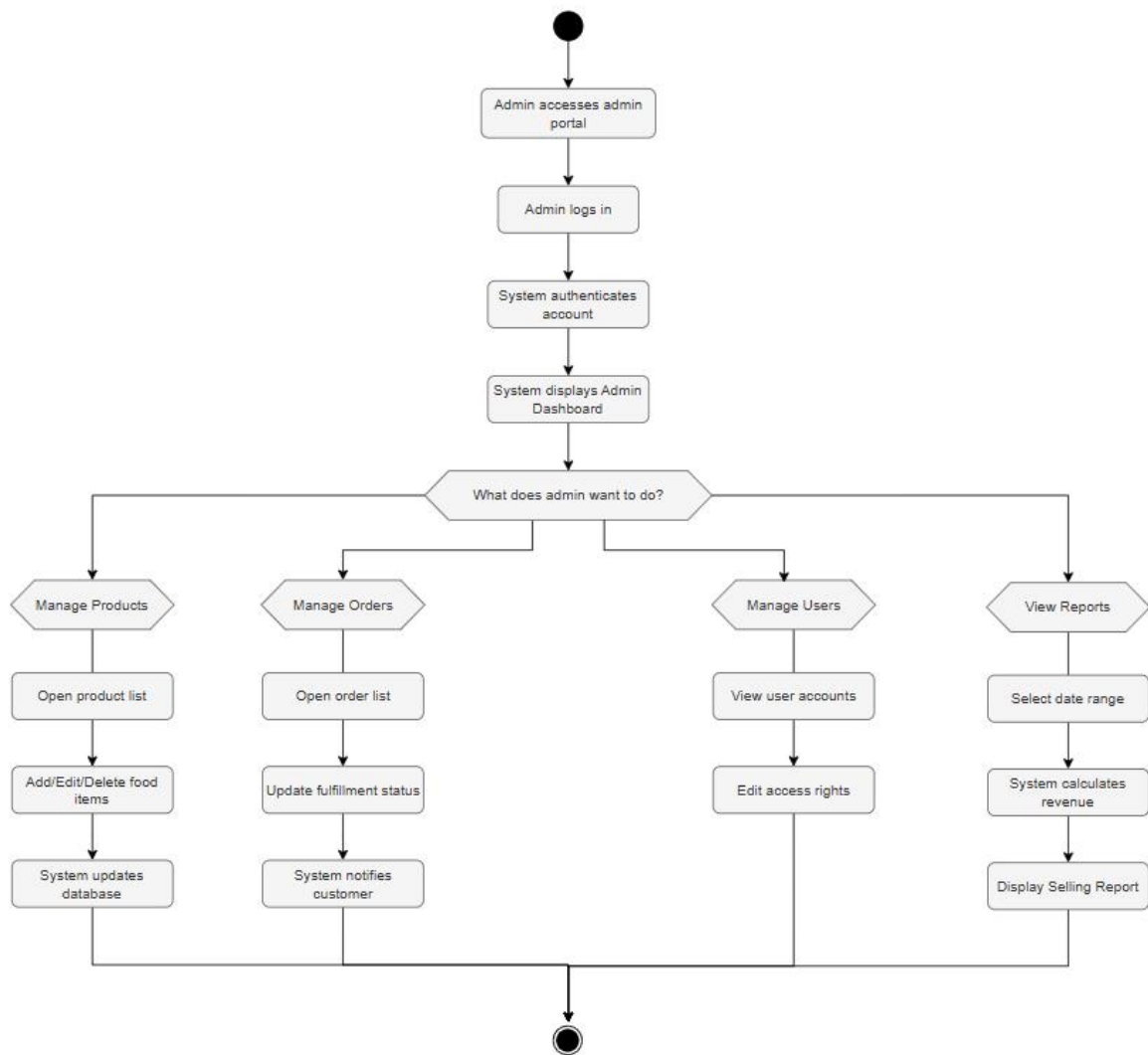


Figure 12 Activity Diagram for Admin Interaction with the System

The diagram shows how an administrator logs into the backend dashboard to oversee the system operations. The workflow includes managing product inventory and categories, updating customer order statuses, managing user accounts, and generating specific timeframe revenue reports.

3.3.3. Baseflow Customer Workflow

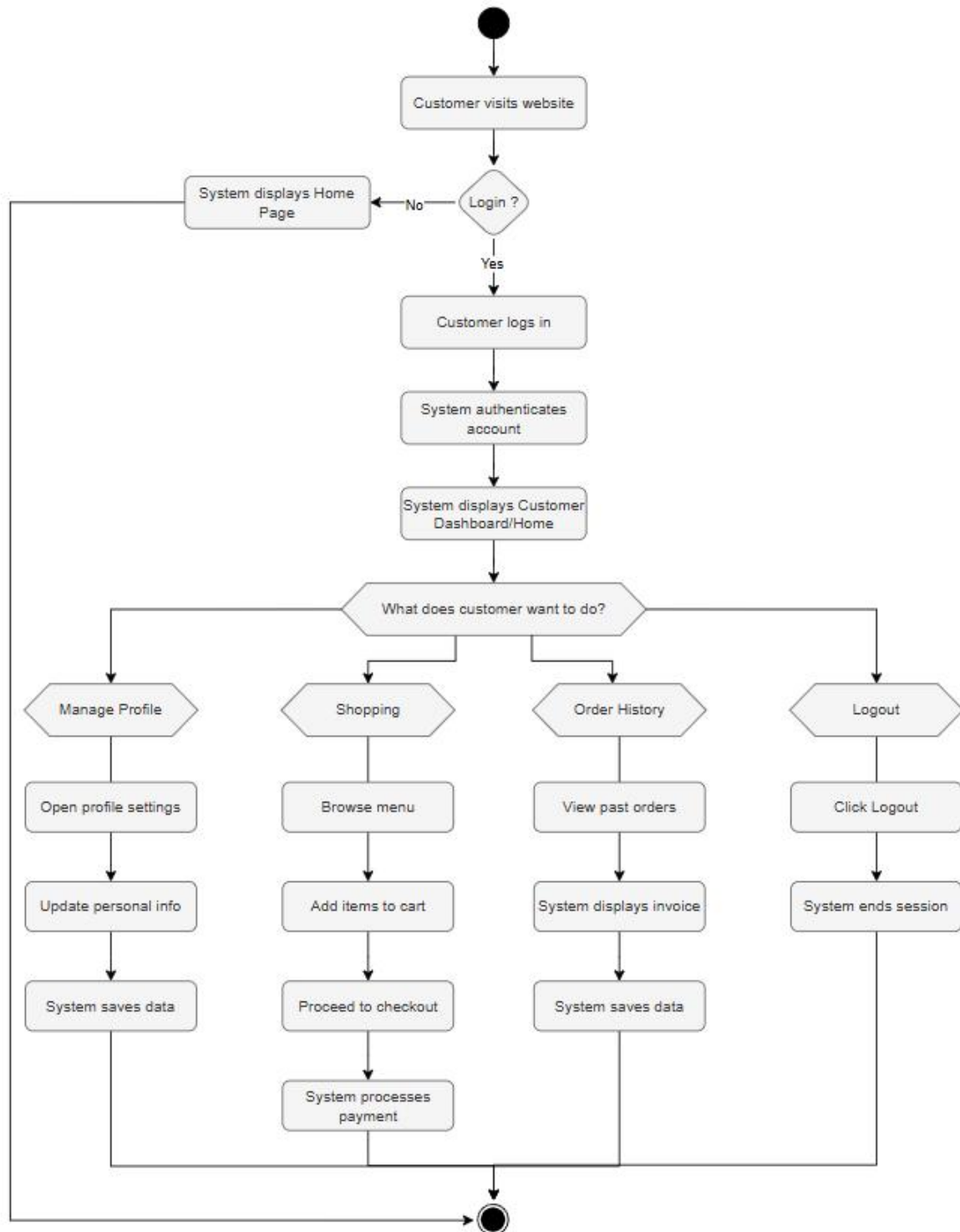


Figure 13 Activity Diagram for Customer Interaction with the System

The diagram shows how a registered customer logs into the system to manage their personal profile, browse food items, add products to their shopping cart, proceed with the checkout process, view their order history, and download invoices.

3.4. Database

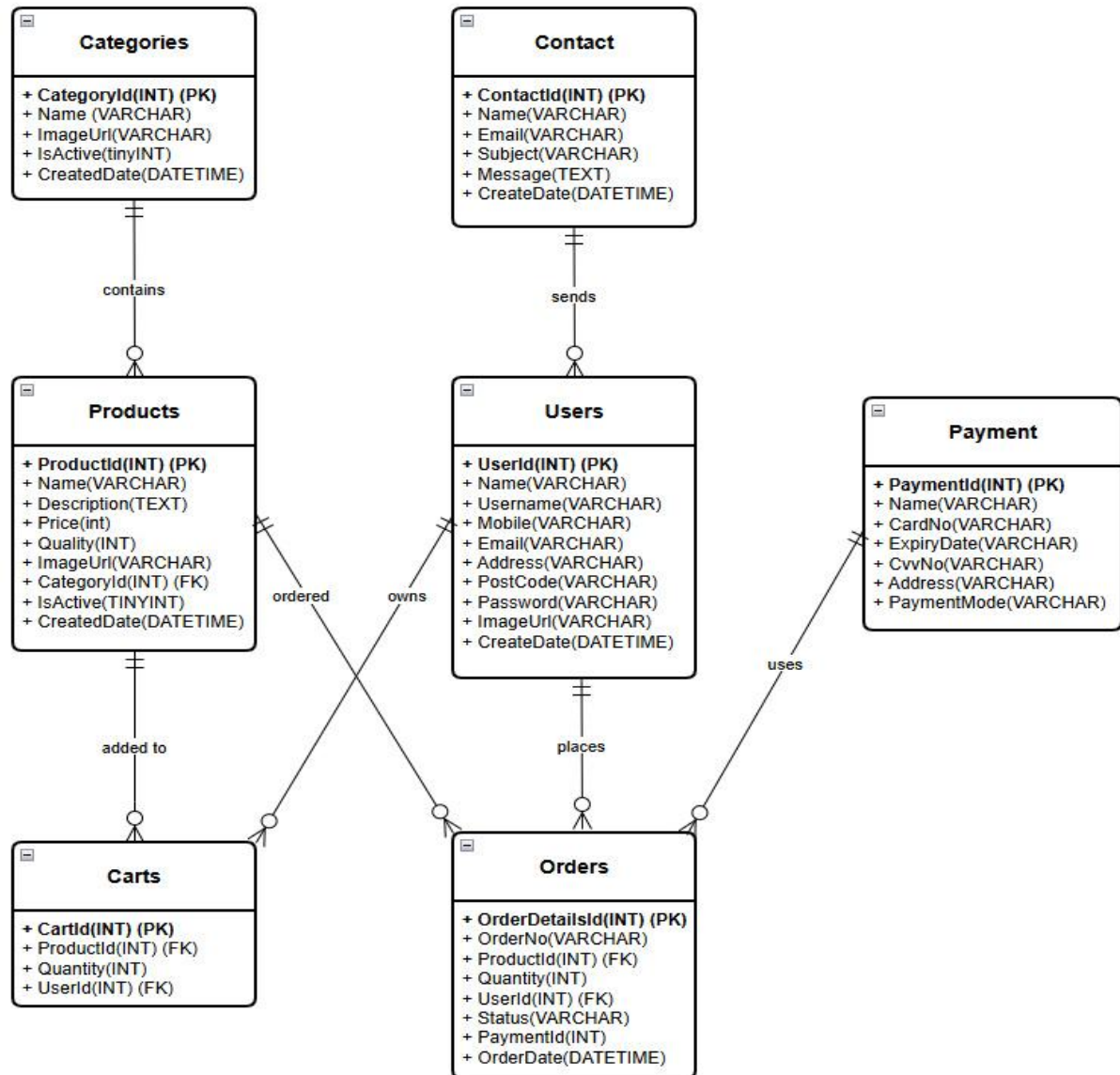


Figure 14 Relationship Diagram

3.4.1. User's Table

Table 1 User's Table

STT	Field Name	Data Type	Description
1	UserId	INT (Primary Key)	Unique identifier for each user
2	Name	VARCHAR	User's full name
3	Username	VARCHAR	Login username
4	Mobile	VARCHAR	Contact phone number
5	Email	VARCHAR	User's email address
6	Address	VARCHAR	User's residential address
7	PostCode	VARCHAR	Postal code

8	Password	VARCHAR	User's encrypted password
9	ImageUrl	VARCHAR	Profile image path
10	CreatedDate	DATETIME	Account creation date

3.4.2. Categories Table

Table 2 Categories Table

STT	Field Name	Data Type	Description
1	CategoryId	INT (Primary Key)	Unique identifier for each category
2	Name	VARCHAR	Category name
3	ImageUrl	VARCHAR	Category image
4	IsActive	TINY INT	Category status
5	CreatedDate	DATETIME	Creation Date

3.4.3. Products Table

Table 3 Product Table

STT	Field Name	Data Type	Description
1	ProductId	INT (Primary Key)	Unique identifier for each product
2	Name	VARCHAR	Product name
3	Description	TEXT	Product description
4	Price	INT	Product price
5	Quantity	INT	Available stock quantity
6	ImageUrl	VARCHAR	Product image
7	CategoryId	INT (Foreign Key)	Related category identifier
8	IsActive	TINY INT	Product status
9	CreatedDate	DATETIME	Creation Date

3.4.4. Carts Table

Table 4 Cart Table

STT	Field Name	Data Type	Description
1	CartId	INT (Primary Key)	Unique identifier for each cart item

2	ProductId	INT (Foreign Key)	Selected product identifier
3	Quantity	INT	Quantity of the selected product
4	UserId	INT (Foreign Key)	Owner of the cart item

3.4.5. Orders Table

Table 5 Order Table

STT	Field Name	Data Type	Description
1	OrderDetailsId	INT (Primary Key)	Unique identifier for each order record
2	OrderNo	VARCHAR	Order number
3	ProductId	INT (Foreign Key)	Purchased product identifier
4	Quantity	INT	Quantity ordered
5	UserId	INT (Foreign Key)	Customer identifier
6	Status	VARCHAR	Payment method identifier
7	PaymentId	INT (Foreign Key)	Order status
8	OrderDate	DATETIME	Date of order placement

3.4.6. Payment Table

Table 6 Payment Table

STT	Field Name	Data Type	Description
1	PaymentId	INT (Primary Key)	Unique identifier for each payment
2	Name	VARCHAR	Cardholder or payer name
3	CardNo	VARCHAR	Card number
4	ExpiryDate	VARCHAR	Card expiration date
5	CvvNo	VARCHAR	Card verification value
6	Address	VARCHAR	Billing address
7	PaymentMode	VARCHAR	Payment method type

3.4.7. Contact Table

Table 7 Contact Table

STT	Field Name	Data Type	Description
-----	------------	-----------	-------------

1	ContactId	INT (Primary Key)	Unique identifier for each contact message
2	Name	VARCHAR	User who submitted the message
3	Email	VARCHAR	User's email use to contact
4	Subject	VARCHAR	Message subject
5	Message	TEXT	Message content
6	CreateDate	DATETIME	Date the message was submitted

3.4.8. Relationships Between Entities

Table 8 Relationships Between Entities Table

STT	Parent Table	Child Table	Relationship	Foreign Key	Description
1	Categories	Products	1 : N	Products.CategoryId → Categories.CategoryId	One category can contain multiple products.
2	Users	Carts	1 : N	Carts.UserId → Users.UserId	One user can have multiple items in their shopping cart.
3	Products	Carts	1 : N	Carts.ProductId → Products.ProductId	One product can appear in multiple shopping carts.
4	Users	Orders	1 : N	Orders.UserId → Users.UserId	One user can place multiple orders.
5	Products	Orders	1 : N	Orders.ProductId → Products.ProductId	One product can be ordered multiple times.

6	Payment	Orders	1 : N	Orders.PaymentId → Payment.PaymentId	One payment method can be used by multiple orders.
7	Users	Contact	1 : N	Contact.UserId → Users.UserId	One user can submit multiple contact requests or feedback messages.

CHAPTER 4. RESULT

4.1. Login and Sign-up Screen

4.1.1. Login Screen

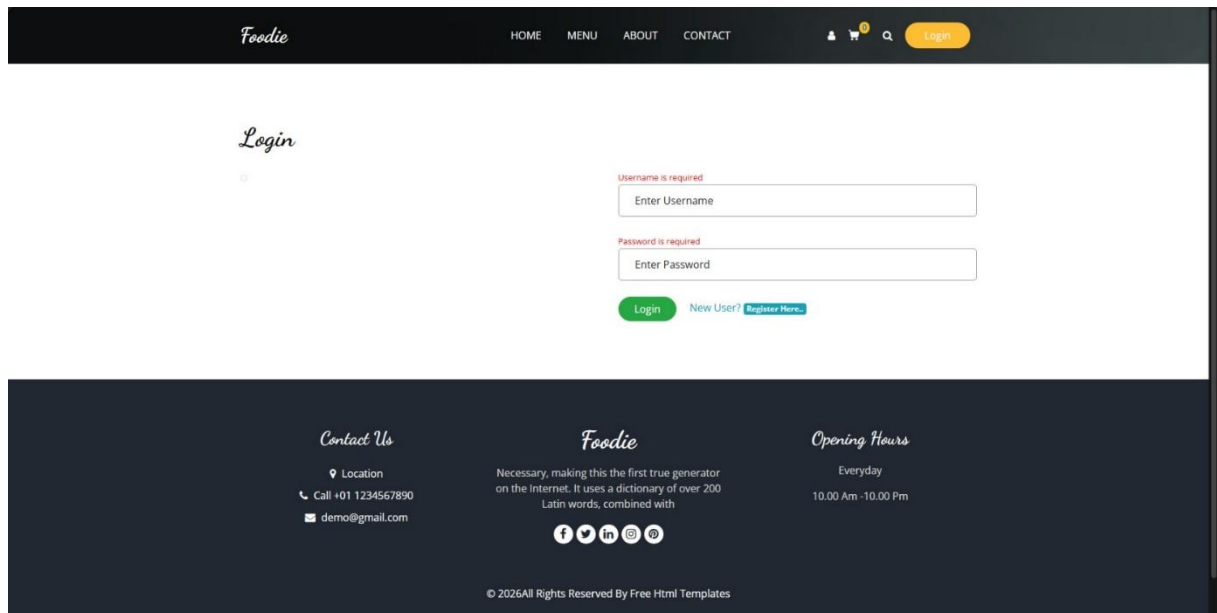
The screenshot shows the 'Foodie' login page. At the top is a dark navigation bar with the 'Foodie' logo, links for HOME, MENU, ABOUT, and CONTACT, and icons for a user profile, shopping cart, and search. The main content area has a light background with the word 'Login' in a script font. Below it are two input fields: 'Enter Username' and 'Enter Password'. Above the username field is a red error message 'Username is required'. Above the password field is a red error message 'Password is required'. Below the password field are three buttons: a green 'Login' button, a blue 'New User?' link, and a blue 'Register Here.' link. The footer is a dark section with three columns: 'Contact Us' with location, phone, and email; 'Foodie' with a description and social media icons; and 'Opening Hours' with the text 'Everyday 10.00 Am -10.00 Pm'. A copyright notice is at the bottom center.

Figure 15 Login Page

In addition, the interface provides a "Register Here" link that redirects new users to the registration page. The navigation bar is consistently designed across the entire website, allowing users to easily access pages such as Home, Menu, About, and Contact. A footer section is also included, displaying contact information, opening hours, and social media links to enhance the professionalism and usability of the website.

The Login Interface is designed to authenticate users before they can access the system's services and features. This page contains essential input fields such as Username, Password, and a Login button. The system validates user input and displays appropriate error messages whenever required fields are left blank, helping users provide accurate information.

Main Functions:

- + Authenticate user accounts.
- + Validate login information.
- + Display error messages for missing required fields.
- + Provide navigation to the registration page.

4.1.2. Register Screen

Foodie HOME MENU ABOUT CONTACT [Login](#)

User Registration

Enter Full Name

Enter Username Username is required

Enter Email

Enter Mobile Number

Enter Address

Enter Post/Zip Code

[Choose File](#) No file chosen

Enter Password Password is required

[Register](#) [Already registered? Login here.](#)

Contact Us
 Location
 Call +01 1234567890
 demo@gmail.com

Foodie
Necessary, making this the first true generator on the Internet. It uses a dictionary of over 200 Latin words, combined with

Opening Hours
Everyday
10.00 Am -10.00 Pm

Figure 16 Register Page

The User Registration Interface allows new users to create an account and become registered members of the system. The registration form collects various personal details, including:

- + Full Name
- + Username
- + Email Address
- + Mobile Number
- + Address
- + Post/Zip Code
- + Profile Image Upload
- + Password

The system performs input validation to ensure that all required fields are completed correctly. Error messages are displayed whenever invalid or incomplete information is submitted. After successful registration, users can use their newly created credentials to log in to the system.

The registration page is designed using a two-column layout, providing a clean and organized user experience. A "Login Here" link is also available for users who already have an account and wish to return to the login page.

Main Functions:

- + Create new user accounts.
- + Collect and store user information.

- + Support profile image uploads.
- + Validate user input data.
- + Provide direct navigation to the login page.

4.2. Main Page

4.2.1. Home Page

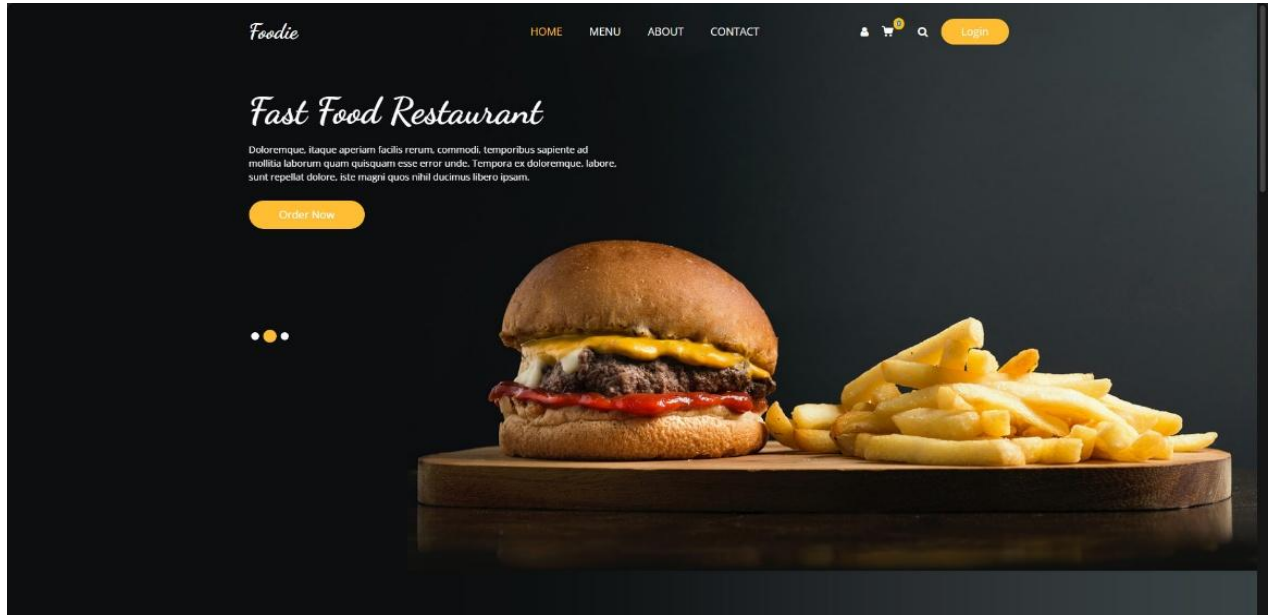
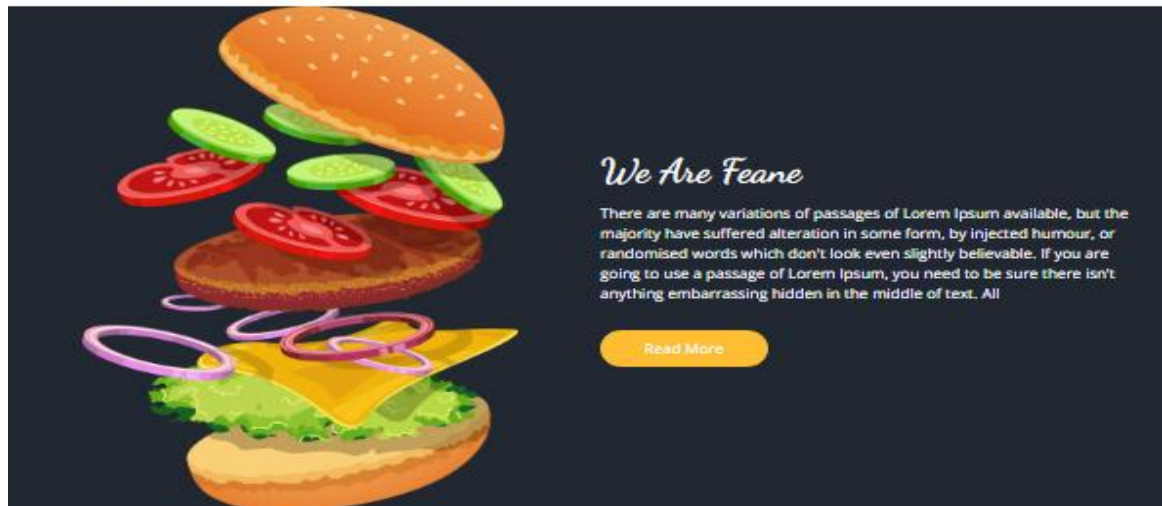
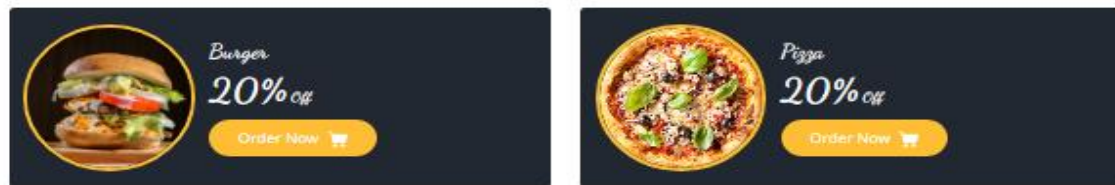


Figure 17 Home Page



What Says Our Customers

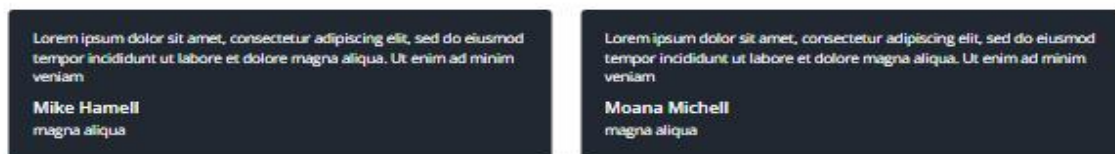


Figure 18 Home Page

The Home Page serves as the main landing page of the Foodie website, providing users with an overview of the restaurant and its services. The page features a modern navigation bar containing links to Home, Menu, About, and Contact pages, along with user account, shopping cart, search, and login functions. A prominent hero section showcases featured food items and promotional content, accompanied by an "Order Now" button to

encourage customer engagement. Below the hero section, promotional banners highlight discounted products, while the About Us section introduces the restaurant and its offerings. The homepage also includes a Customer Testimonials section that displays customer reviews to build trust and credibility. Finally, the footer section provides essential information such as contact details, social media links, and opening hours. Overall, the Home Page is designed to be visually appealing, user-friendly, and effective in promoting the restaurant's products and services while guiding visitors toward placing orders.

4.2.2. Food Menu

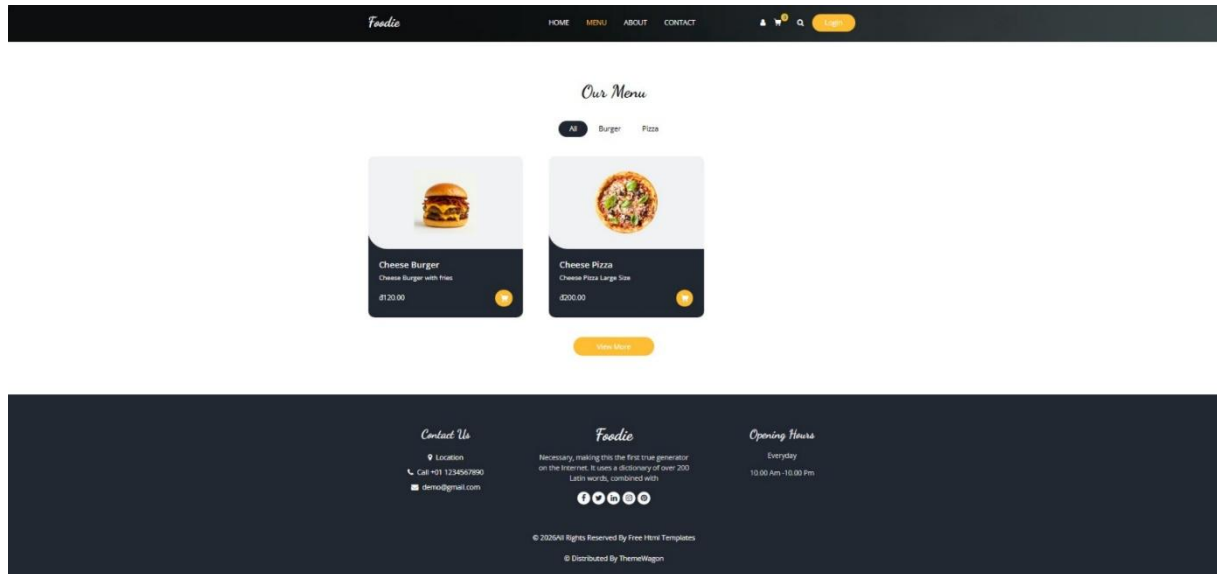


Figure 19 Food Menu Page

The Food Menu Page allows customers to browse and explore the restaurant's available food items. The page presents products in a clean card-based layout, displaying essential information such as product images, names, descriptions, and prices. Category filters, including All, Burger, and Pizza, enable users to quickly view products based on their preferences. Each menu item includes an Add to Cart button, allowing customers to conveniently add products to their shopping cart for future purchase. A View More button is also provided to load additional menu items when available. Like other pages in the system, the Food Menu Page includes a navigation bar for easy access to website features and a footer containing contact information, social media links, and opening hours. Overall, this page enhances the user experience by providing an organized and efficient way to browse menu items and place orders.

4.2.3. About Us Page

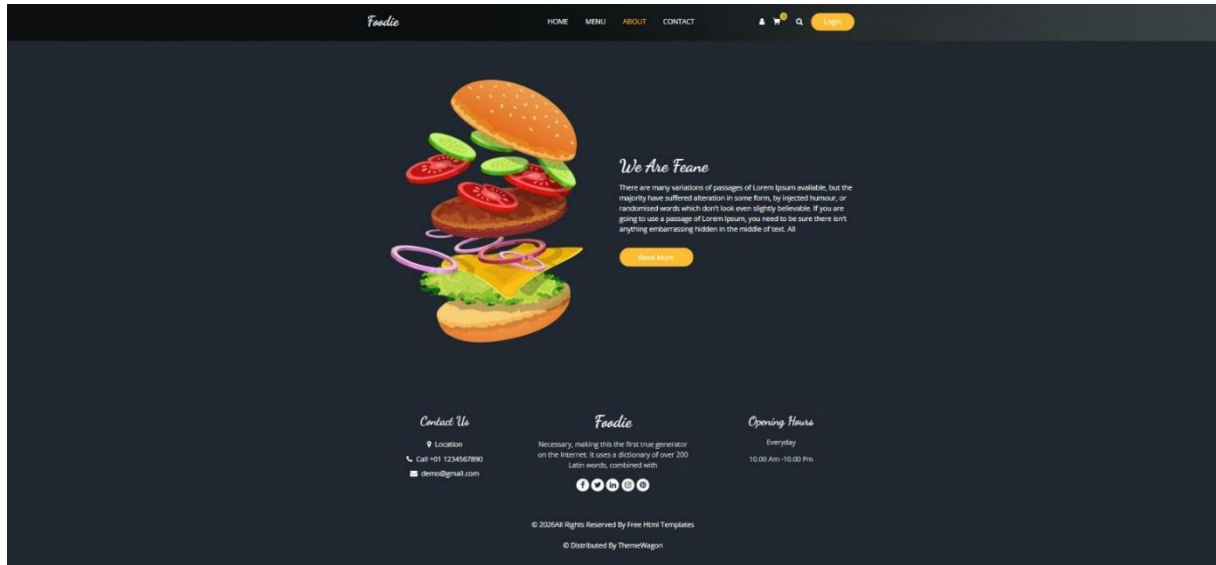


Figure 20 About Us Page

The About Us Page provides visitors with information about the Foodie restaurant, its background, and its commitment to delivering high-quality food and services. The page features an attractive food illustration alongside descriptive content that introduces the restaurant and its values. A "Read More" button is included to allow users to access additional information when needed. The page maintains a consistent design with the rest of the website through the use of the same navigation bar, color scheme, typography, and footer section. Contact information, social media links, and opening hours are displayed at the bottom of the page to help customers easily reach the restaurant. Overall, the About Us Page helps build customer trust, strengthen the restaurant's brand identity, and provide visitors with a better understanding of the business.

4.2.4. Contact Page

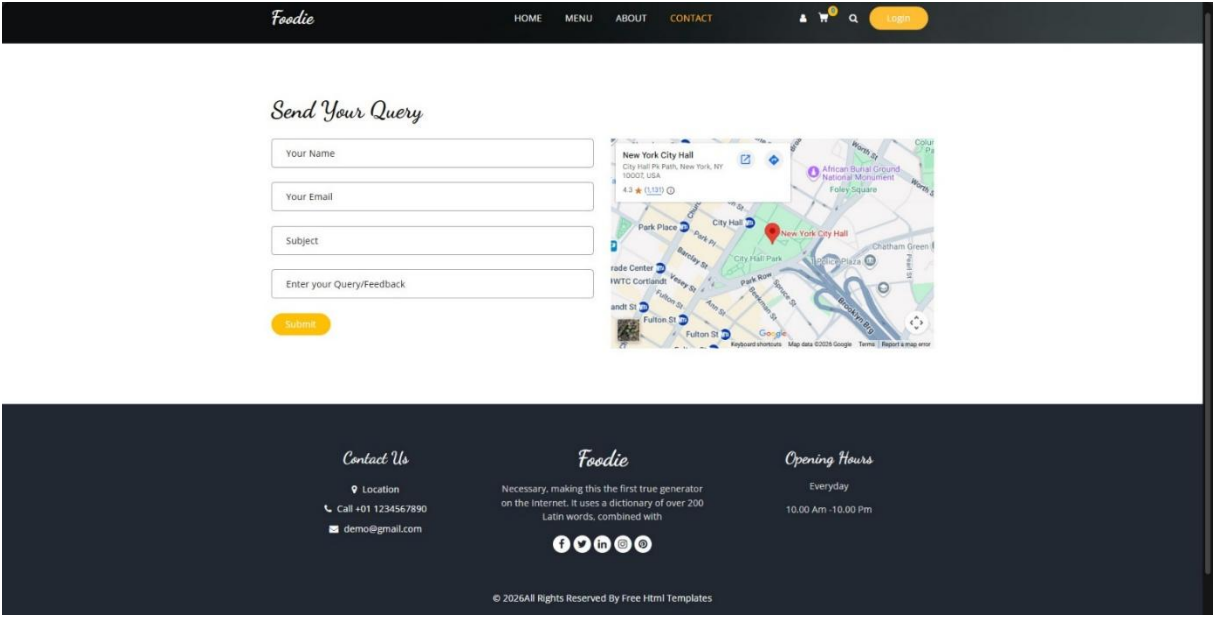


Figure 21 Contact Form Page

The Contact Page enables customers to communicate directly with the restaurant by submitting inquiries, feedback, or support requests. The page includes a contact form with fields for Name, Email, Subject, and Message, allowing users to provide detailed information regarding their inquiries. A Submit button is provided to send the entered information to the system for processing. Additionally, an embedded Google Map displays the restaurant's location, helping customers easily find directions and access the physical store. Consistent with the overall website design, the page includes a navigation bar for easy access to other sections and a footer containing contact details, social media links, and opening hours. Overall, the Contact Page enhances customer communication and improves accessibility by providing multiple ways for users to reach the restaurant.

4.2.5. User Profile Page

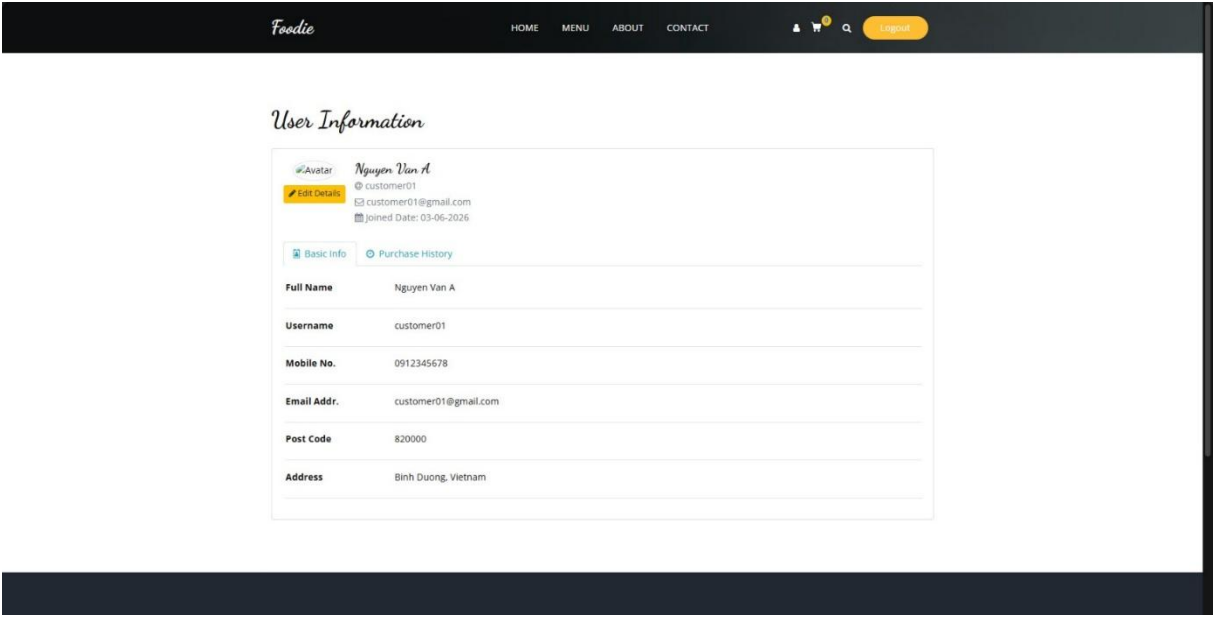


Figure 22 User Profile Page

The User Profile Page allows registered users to view and manage their personal information within the Foodie website. The page displays essential account details, including the user's profile picture, full name, username, email address, mobile number, postal code, address, and account creation date. An Edit Details button is provided to enable users to update their profile information whenever necessary. Additionally, the interface includes tabs such as Basic Info and Purchase History, allowing users to easily switch between personal information and their previous orders. The page follows the same design style as the rest of the website, maintaining a consistent navigation bar and footer for a seamless user experience. Overall, the User Profile Page enhances user convenience by providing a centralized location for account management and purchase tracking.

User Information

Avatar

Nguyen Van A

@customer01

customer01@gmail.com

Joined Date: 10-10-2028

Basic Info

Purchase History

Payment Mode: COD Card No: 123456789 Invoice					
Product Name	Unit Price	Qty	Total Price	Order No	Status
Cheese Burger	\$120.00	2	\$240.00	OD001	Pending
Cheese Pizza	\$200.00	1	\$200.00	OD002	Delivered
			\$440		

Payment Mode: COD Card No: 123456789 Invoice					
Product Name	Unit Price	Qty	Total Price	Order No	Status
Cheese Burger	\$120.00	2	\$240.00	OD001	Pending
Cheese Pizza	\$200.00	1	\$200.00	OD002	Delivered
			\$440		

Payment Mode: COD Card No: 123456789 Invoice					
Product Name	Unit Price	Qty	Total Price	Order No	Status
Cheese Pizza	\$200.00	1	\$200.00	FC29AC38	Delivered
Cheese Burger	\$120.00	1	\$120.00	B3C7D05E	Pending
			\$320		

Payment Mode: COD Card No: 123456789 Invoice					
Product Name	Unit Price	Qty	Total Price	Order No	Status
Cheese Pizza	\$200.00	1	\$200.00	FC29AC38	Delivered
Cheese Burger	\$120.00	1	\$120.00	B3C7D05E	Pending
			\$320		

Tính năng tìm và đơn hàng đang được cập nhật...

Figure 23 Purchase History Page

The Purchase History section allows users to view and track all previous orders placed through the Foodie website. This page displays detailed information for each transaction, including the payment method, product name, unit price, quantity, total price, order number, and current order status. Status indicators such as Pending and Delivered help users monitor the progress of their orders in real time. Additionally, an Invoice button is provided for each order, enabling users to view or download transaction details when needed. Integrated within the User Profile page, the Purchase History feature provides customers with a convenient way to review past purchases, track order fulfillment, and maintain a record of their transactions. Overall, this functionality improves transparency and enhances the user experience by offering easy access to order-related information.

4.2.6. Cart Page

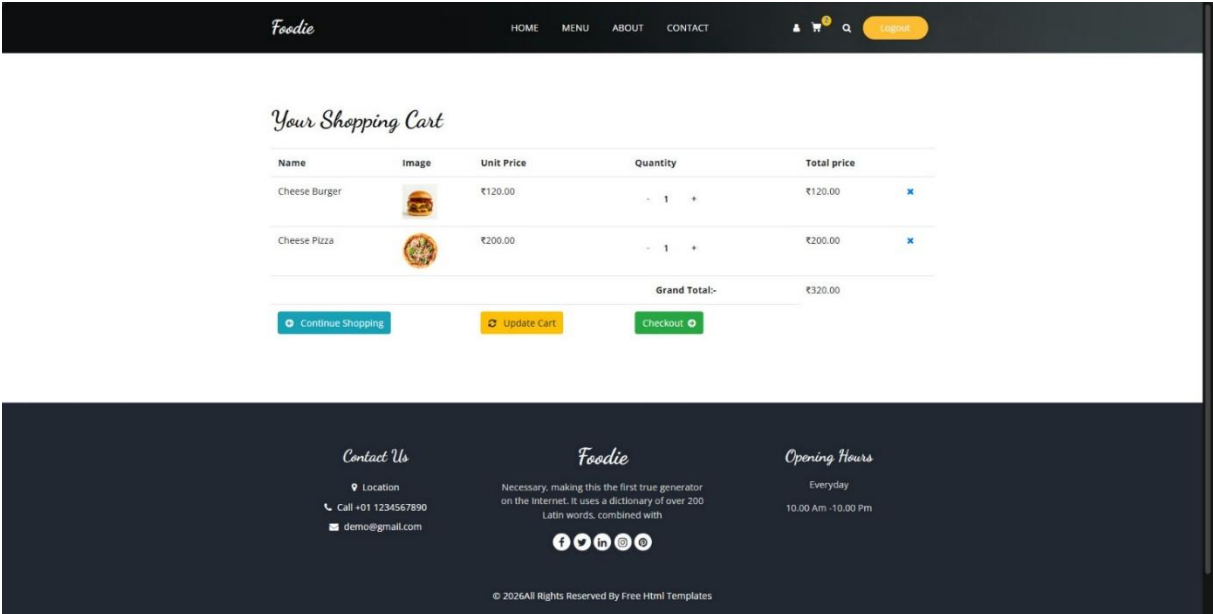
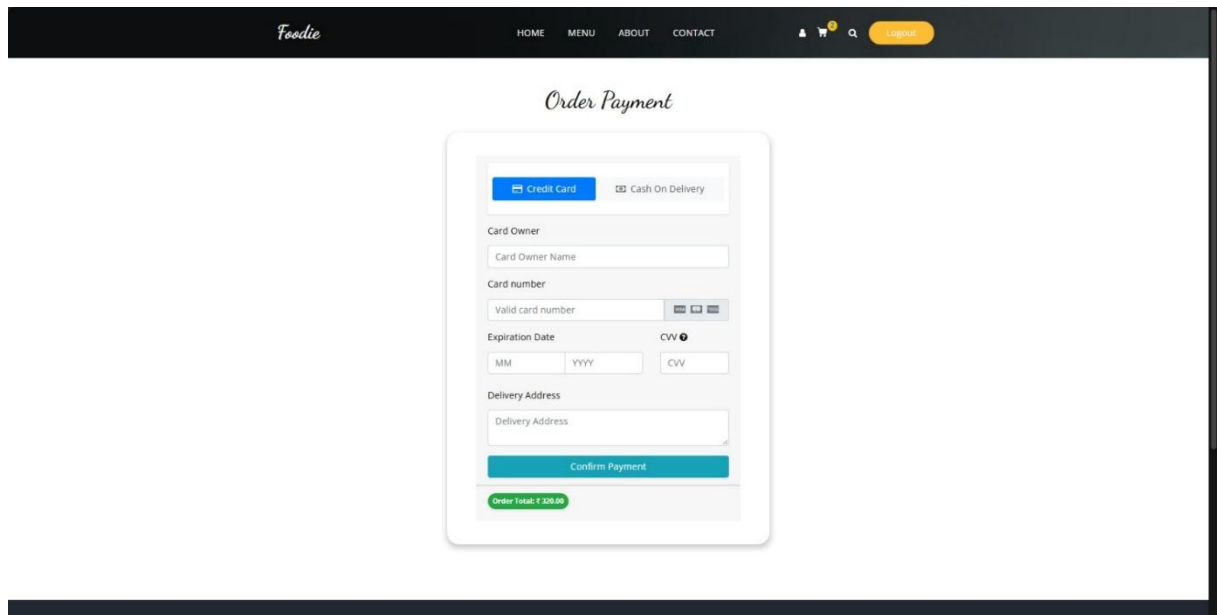


Figure 24 Cart Page

The Shopping Cart Page allows users to review and manage the products they intend to purchase before proceeding to checkout. The page displays detailed information for each selected item, including the product name, image, unit price, quantity, and total price. Users can easily increase or decrease item quantities, remove products from the cart, and update the cart to reflect any changes. The system automatically calculates and displays the Grand Total of all items in the cart. Additionally, the page provides action buttons such as Continue Shopping, Update Cart, and Checkout, enabling users to continue browsing products, modify their orders, or proceed to the payment process. By providing a clear summary of selected items and order costs, the Shopping Cart Page helps users make informed purchasing decisions and improves the overall online ordering experience.

4.2.7. Payment Page



The screenshot displays the 'Order Payment' page for a website named 'Foodie'. The page features a dark header with navigation links (HOME, MENU, ABOUT, CONTACT), a shopping cart icon, and a 'Logout' button. The main content area is titled 'Order Payment' and contains a payment form. The form has two tabs: 'Credit Card' (selected) and 'Cash On Delivery'. The 'Credit Card' tab includes fields for 'Card Owner Name', 'Card number' (with a 'Valid card number' label and a card icon), 'Expiration Date' (split into 'MM' and 'YYYY' fields), 'CVV' (with a 'CVV' label and a card icon), and 'Delivery Address'. A 'Confirm Payment' button is located at the bottom of the form. Below the button, a green box displays the 'Order Total: ₹ 500.00'.

Figure 25 Payment (Credit Card Method)

The Payment Page allows customers to securely complete their orders using a credit card payment method. The interface provides a structured payment form where users are required to enter essential billing information, including the cardholder name, card number, expiration date, CVV code, and delivery address. The page also offers multiple payment options, allowing customers to switch between Credit Card and Cash on Delivery (COD) according to their preference. A Confirm Payment button is provided to finalize the transaction, while the total order amount is clearly displayed to ensure transparency before payment submission. The simple and user-friendly design helps streamline the checkout process, improving convenience and enhancing the overall online ordering experience.

Foodie HOME MENU ABOUT CONTACT 🛒 🔍 📄 [Logout](#)

Order Payment

☐ Credit Card
 ☒ Cash On Delivery

Delivery Address

123, Khu Phố A, phường B, thành phố C

[Confirm Order](#)

Note: At the time of receiving your order, you need to make full payment. After completing the delivery process, you can check your updated order history status.

Order Total: \$ 220.00

Contact Us

📍 Location

☎ Call +01 1234567890

✉ demo@gmail.com

Foodie

Necessary, making this the first true generator on the Internet. It uses a dictionary of over 200 Latin words, combined with

Opening Hours

Everyday

10.00 Am -10.00 Pm

Figure 26 Payment (Cash of Delivery Method)

The Cash on Delivery (COD) Payment Page allows customers to place orders and pay upon receiving their products. Unlike the credit card payment option, this method only requires users to provide a delivery address before confirming the order. The page includes a Cash on Delivery payment option, a delivery address input field, and a Confirm Order button to complete the purchase process. Additional information is displayed to inform customers that payment will be collected when the order is delivered and that order status updates can be tracked through the purchase history section. The total order amount is clearly shown to ensure transparency before order confirmation. This payment method provides a simple and convenient checkout experience for customers who prefer to pay in cash upon delivery.

4.2.8. Order Success Page

Foodie HOME MENU ABOUT CONTACT 🛒 🔍 📄 [Logout](#)

ORDER INVOICE

S.No	Order No	Product Name	Price	Qty	Total Price
1	FC256C5E	Cheese Pizza	\$200.00	1	\$200.00
2	B3C7D05E	Cheese Burger	\$120.00	1	\$120.00
Grand Total:					\$320.00

[Download Invoice \(PDF\)](#)
[Print Invoice](#)

Contact Us

📍 Location

☎ Call +01 1234567890

✉ demo@gmail.com

Foodie

Necessary, making this the first true generator on the Internet. It uses a dictionary of over 200 Latin words, combined with

[f](#)
[t](#)
[in](#)
[@](#)

Opening Hours

Everyday

10.00 Am -10.00 Pm

© 2026All Rights Reserved By Free Html Templates

© Distributed By ThemeWagon

Figure 27 Order Information Page

The Order Invoice Page provides customers with a detailed summary of their completed orders. The page displays important transaction information, including the order number, product name, unit price, quantity, and total price for each purchased item. A Grand Total section is included to clearly show the overall amount payable for the order. To improve convenience and record keeping, the system offers options to Download Invoice (PDF) and Print Invoice, allowing customers to save or print their purchase records whenever needed. The page follows the same design style as the rest of the Foodie website, maintaining consistency through the navigation bar and footer components. Overall, the Order Invoice Page enhances transparency, supports transaction verification, and provides users with an accessible record of their purchases.

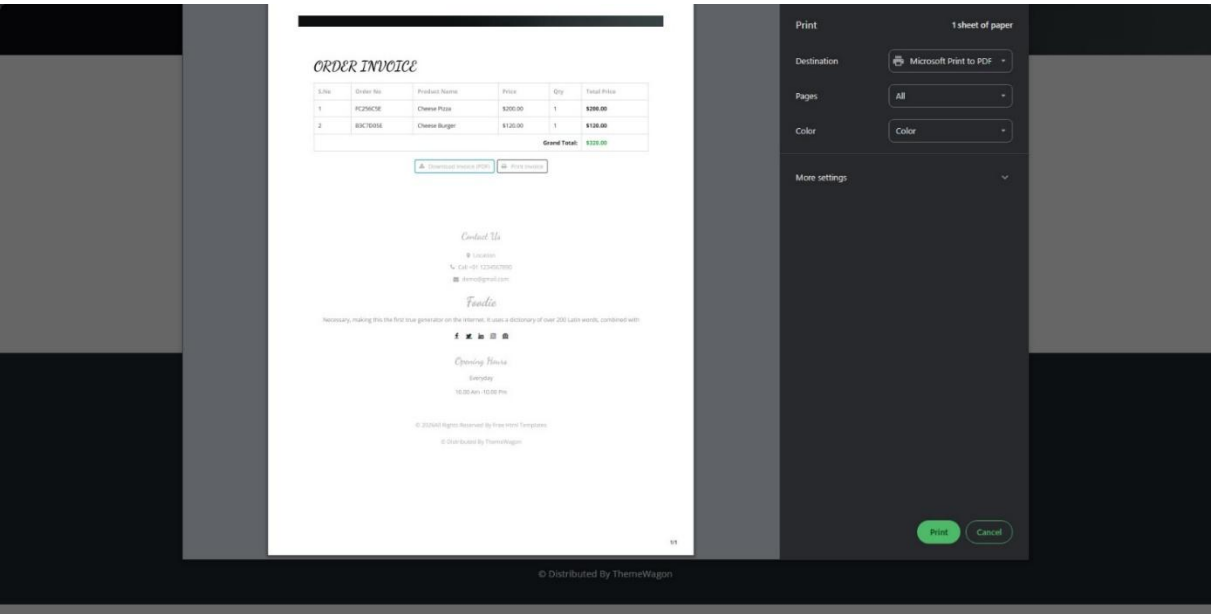


Figure 28 Print the Order Mode

The Print Invoice Interface allows customers to generate a printer-friendly version of their order invoice for documentation and record-keeping purposes. When the Print Invoice button is selected, the system opens the browser's print preview window, displaying a formatted invoice containing order details such as order number, product information, quantity, price, and grand total. Users can choose a printer destination, customize printing settings, or save the invoice as a PDF file using the Microsoft Print to PDF option. This functionality ensures that customers can easily obtain a physical or digital copy of their transaction records for future reference. By providing invoice printing capabilities, the system improves usability, enhances transparency, and supports efficient order management.

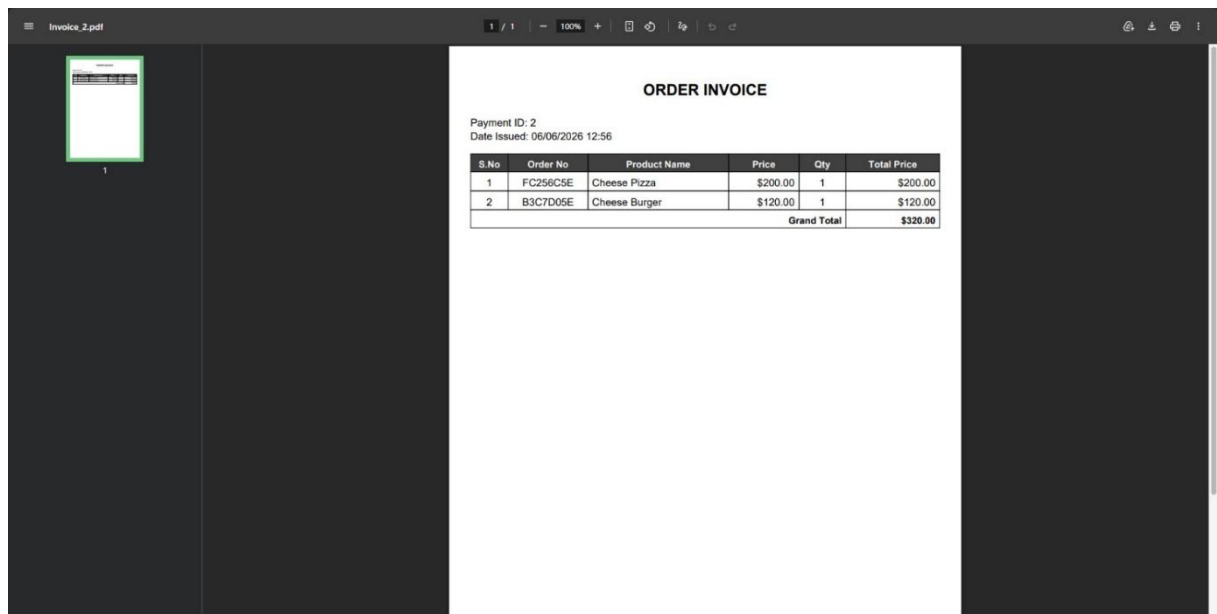


Figure 29 Watch PDF Order Invoice

The PDF Invoice Generation Interface enables customers to download a digital copy of their order invoice in PDF format. After selecting the Download Invoice (PDF) option, the system automatically generates a structured invoice document containing important transaction details, including the payment ID, issue date, order number, product information, quantity, unit price, and grand total. The generated PDF can be viewed directly in the browser, saved to the user's device, or printed for future reference. This feature provides customers with a permanent and portable record of their purchases while improving transaction transparency and documentation. By supporting PDF invoice generation, the Foodie website enhances professionalism, simplifies record management, and improves the overall customer experience.

4.3. Admin Page

4.3.1. Admin Dashboard

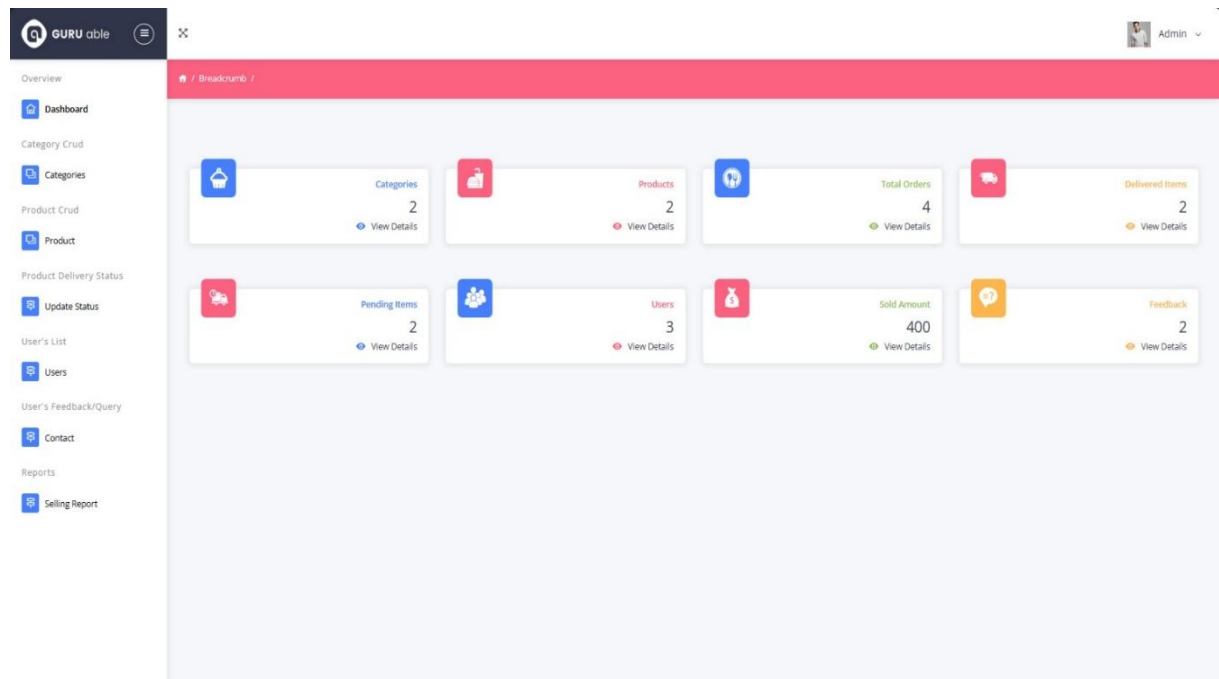


Figure 30 Admin Dashboard

The Admin Dashboard serves as the primary control center of the Foodie website, providing administrators with a comprehensive overview of the system's operational status. The dashboard displays key performance indicators through summary cards, including the total number of categories, products, orders, delivered items, pending orders, registered users, customer feedback records, and total sales revenue. By presenting this information in a visual and organized manner, administrators can quickly assess business performance and monitor daily activities without navigating through multiple pages.

In addition to system statistics, the dashboard provides access to various administrative functions through the navigation panel located on the left side of the interface. Administrators can manage product categories, products, delivery status updates, user accounts, customer inquiries, and sales reports. These management tools enable efficient maintenance of the website and ensure that customer orders, product information, and user data remain accurate and up to date.

The dashboard is designed with a clean and user-friendly layout that enhances usability and supports effective decision-making. By centralizing both analytical information and management functions in a single interface, the Admin Dashboard improves operational efficiency, simplifies administrative tasks, and contributes to the overall success of the online food ordering system.

4.3.2. Categories Management

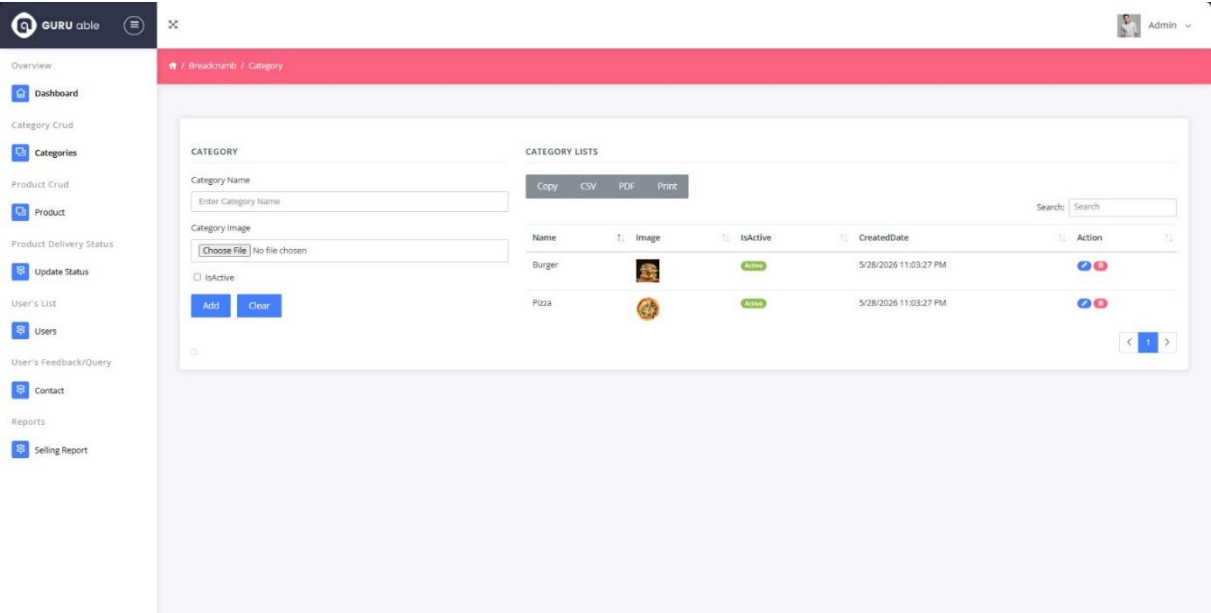


Figure 31 Categories Management

The Category Management Interface allows administrators to create, manage, and organize product categories within the Foodie website. Through this interface, administrators can add new categories by entering a category name, uploading a category image, and setting the active status of the category. The system provides a simple form-based layout that enables administrators to efficiently maintain the product classification structure, ensuring that menu items are grouped correctly and displayed appropriately throughout the website.

In addition to category creation, the interface displays a complete list of existing categories in a searchable and sortable table. Each category record includes information such as category name, image, active status, and creation date. Administrative actions including Edit and Delete are available for each category, allowing easy modification and maintenance of category data. Furthermore, export options such as Copy, CSV, PDF, and Print provide convenient reporting and data management capabilities. By centralizing category administration into a single interface, the Category Management Page improves system organization, simplifies product management, and enhances the overall efficiency of website administration.

4.3.3. Product Management

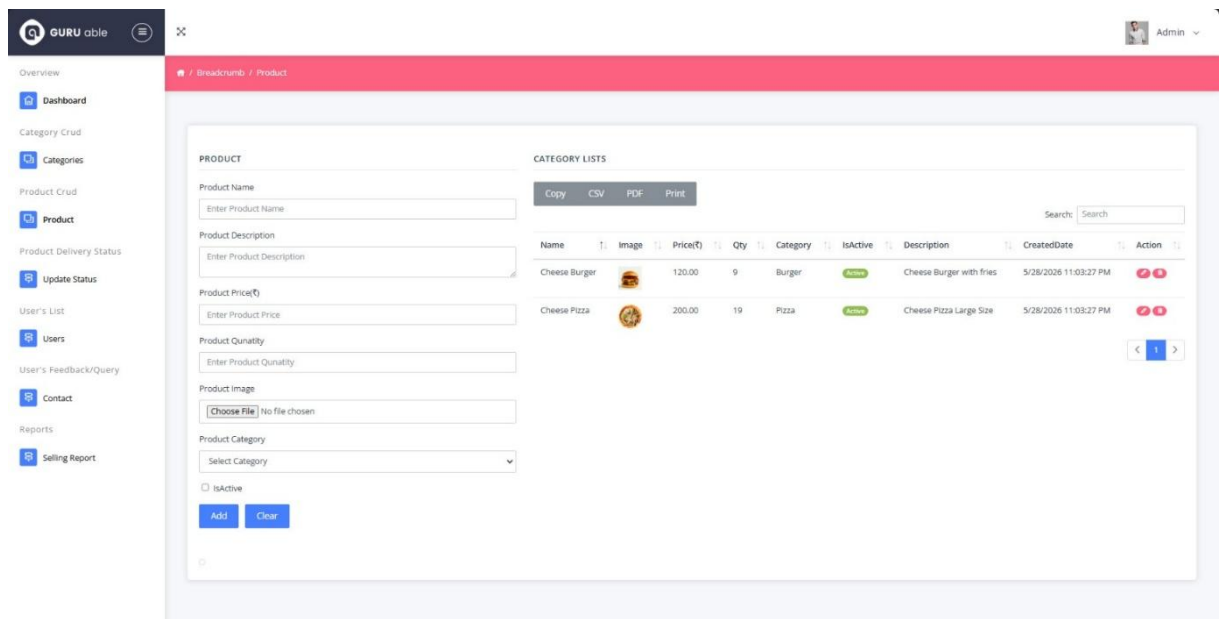


Figure 32 Product Management

The Product Management Interface enables administrators to manage all products available on the Foodie website. Through this interface, administrators can create new products by entering essential information such as product name, description, price, quantity, product image, and category. The system also allows administrators to specify whether a product is active or inactive, ensuring that only available products are displayed to customers. The form-based layout simplifies the process of adding and maintaining product information, making inventory management more efficient and organized.

In addition to product creation, the interface provides a comprehensive product list displayed in a searchable and sortable table. Each record contains detailed information including product name, image, price, quantity, category, status, description, and creation date. Administrators can perform Edit and Delete operations directly from the table, allowing quick updates to product information whenever necessary. Furthermore, data export options such as Copy, CSV, PDF, and Print are available to support reporting and administrative tasks. By providing centralized control over product data, the Product Management Interface helps ensure accurate menu information, effective inventory management, and an improved customer shopping experience.

4.3.4. Order Status Management

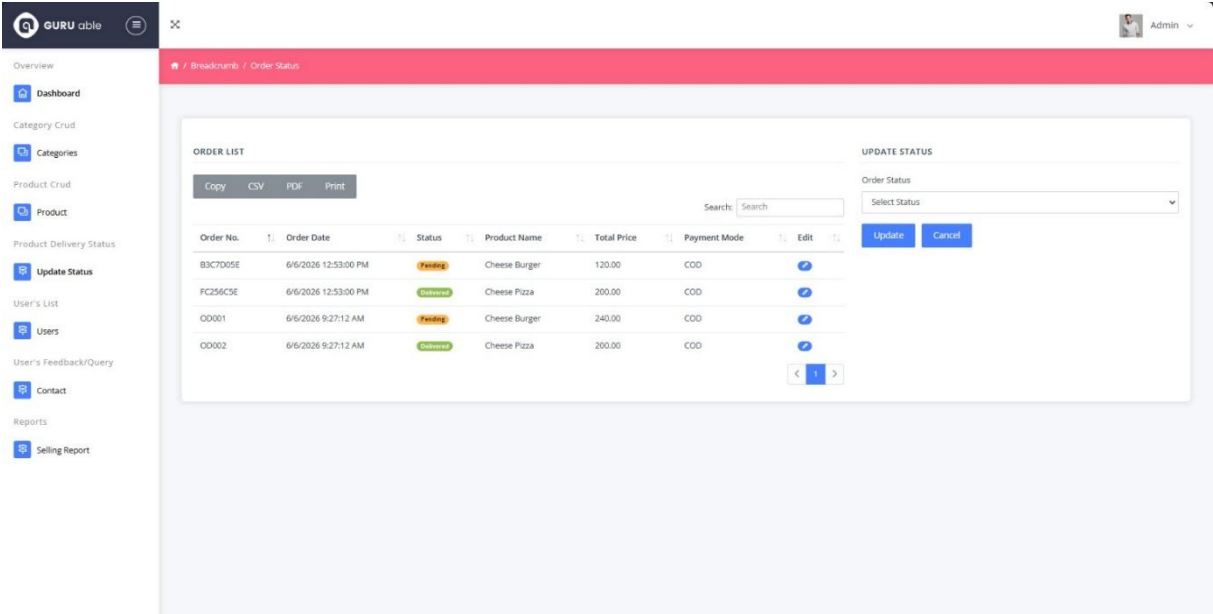


Figure 33 Order's Update Status

The Order Status Management Interface enables administrators to monitor and manage customer orders throughout the order fulfillment process. The page displays a comprehensive list of orders, including important information such as order number, order date, status, product name, total price, and payment method. A searchable and sortable table allows administrators to quickly locate specific orders and review their details. Additionally, export options such as Copy, CSV, PDF, and Print are provided to facilitate reporting and record management.

To ensure efficient order processing, the interface includes an Update Status section where administrators can modify the status of selected orders. Available statuses may include stages such as Pending, Processing, Delivered, or other predefined workflow states. By updating order statuses in real time, administrators can keep customers informed about the progress of their purchases and maintain accurate order records. This functionality improves operational efficiency, enhances customer satisfaction, and supports effective order tracking within the Foodie online food ordering system.

4.3.5. User Management

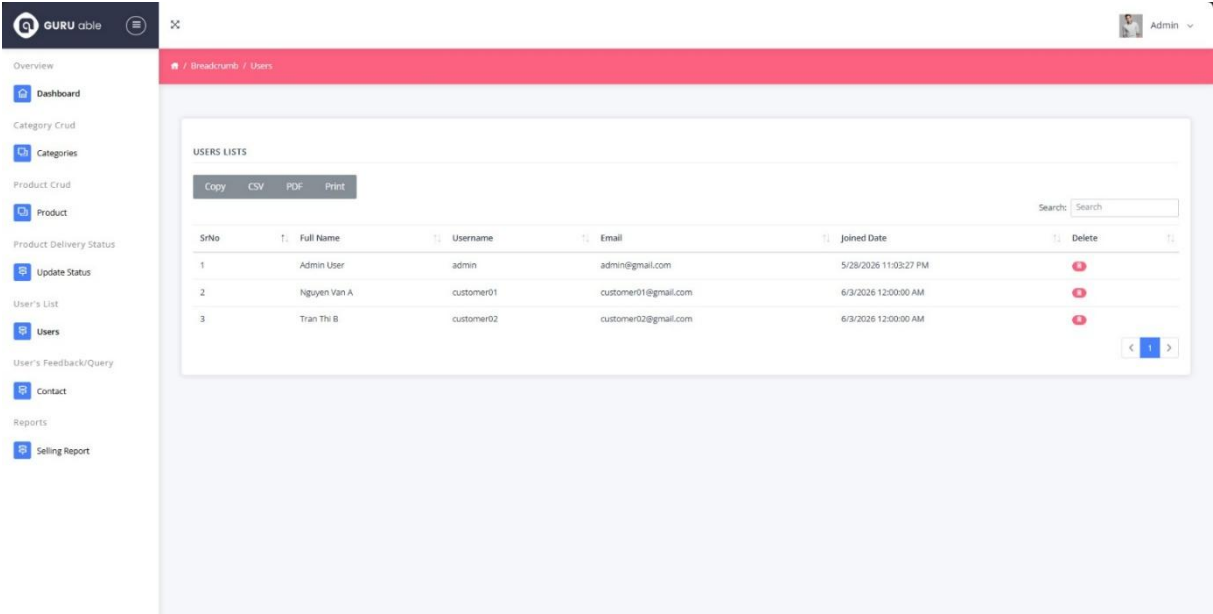


Figure 34 Users Management

The User Management Interface allows administrators to monitor and manage all registered users within the Foodie website. This page provides a centralized view of customer and administrator accounts, displaying important information such as user ID, full name, username, email address, and account registration date. The searchable and sortable table structure enables administrators to quickly locate specific users and review their account information. Additionally, export options including Copy, CSV, PDF, and Print are available to support reporting and administrative documentation requirements.

To maintain the integrity and security of the system, administrators are provided with account management functions such as deleting user accounts when necessary. By offering a comprehensive overview of all registered users, the interface helps administrators effectively manage customer records, monitor user activity, and maintain an organized database. This functionality contributes to improved system administration, better user account control, and enhanced overall management of the Foodie online food ordering platform.

4.3.6. Contact Management

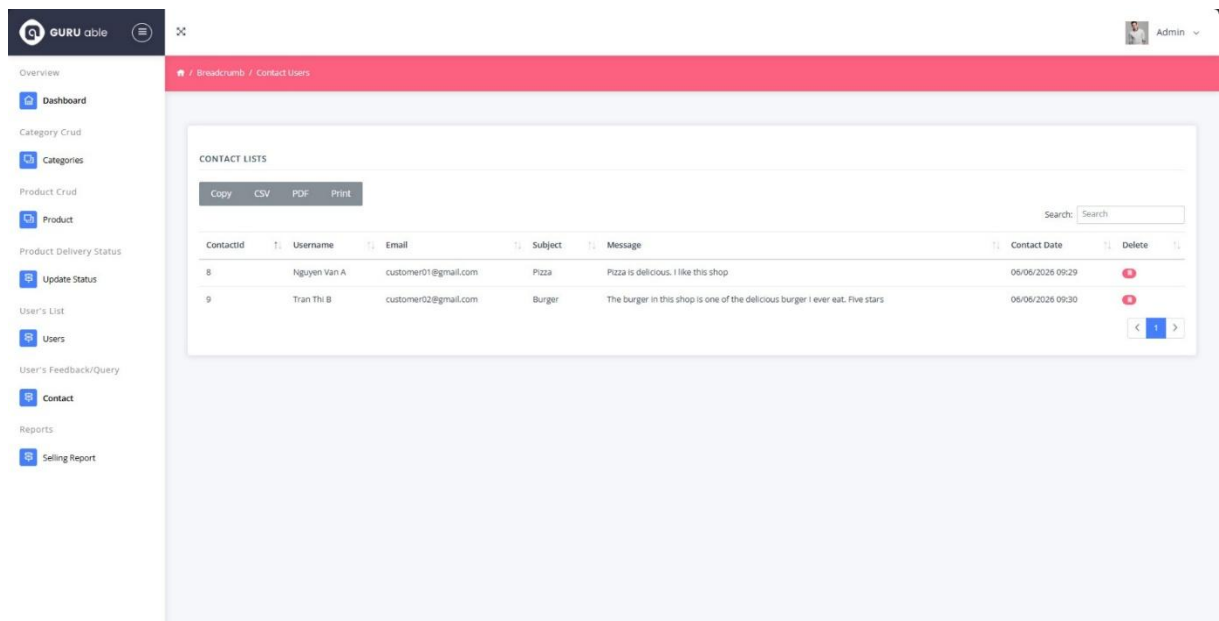


Figure 35 Contact Management

The Contact Management Interface enables administrators to monitor and manage messages submitted by customers through the Contact Page of the Foodie website. This page provides a centralized repository of customer inquiries, feedback, and suggestions, allowing administrators to review user communications efficiently. Each contact record contains important information such as contact ID, username, email address, subject, message content, and submission date. The searchable and sortable table structure helps administrators quickly locate specific messages and organize customer communications effectively.

To support administrative operations and reporting, the interface also includes export functionalities such as Copy, CSV, PDF, and Print. Administrators can remove outdated or irrelevant messages using the delete function, ensuring that the contact database remains organized and manageable. By providing a dedicated platform for handling customer inquiries and feedback, the Contact Management Interface improves communication between customers and the business, supports customer service activities, and helps administrators better understand customer opinions and concerns. This functionality contributes to higher customer satisfaction and more effective management of the Foodie online food ordering system.

4.3.7. Selling Report

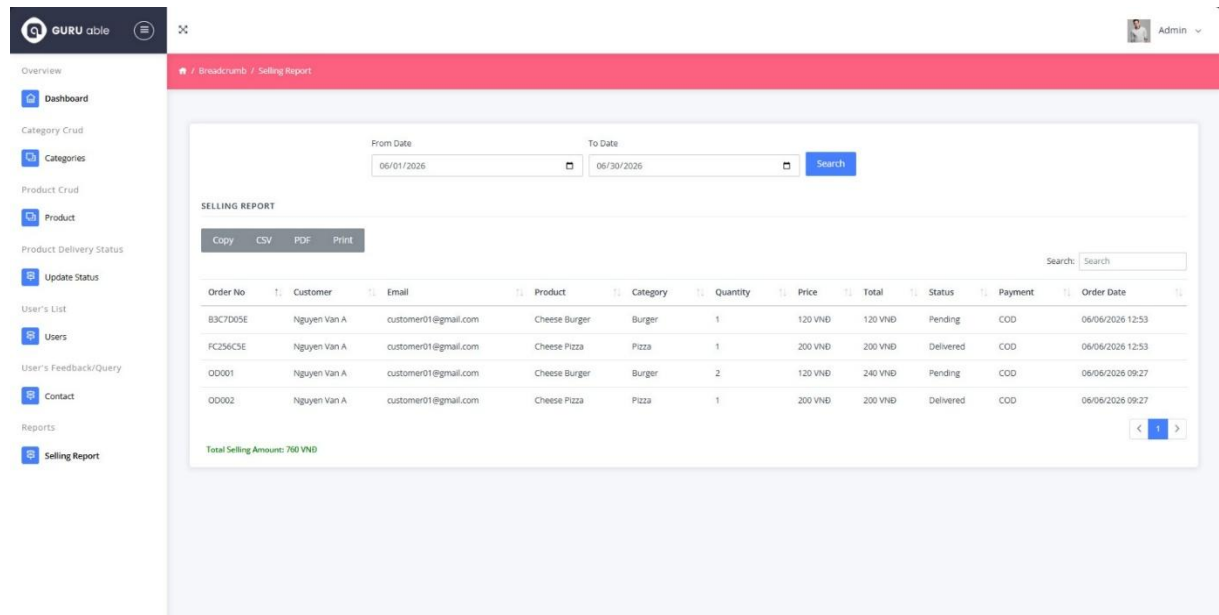


Figure 36 Selling Reports

The Sales Report Interface provides administrators with detailed insights into the sales performance of the Foodie website. This page allows administrators to generate sales reports within a specified date range by selecting a From Date and To Date and executing a search. The generated report displays comprehensive transaction information, including order number, customer name, email address, product name, category, quantity sold, product price, total transaction amount, payment method, order status, and order date. This information enables administrators to monitor sales activities and evaluate business performance over a selected period.

To facilitate reporting and data analysis, the interface includes export options such as Copy, CSV, PDF, and Print, allowing administrators to save or share sales records in multiple formats. The page also calculates and displays the Total Selling Amount, providing a quick overview of total revenue generated during the selected reporting period. By offering accurate sales statistics and flexible reporting capabilities, the Sales Report Interface supports business decision-making, financial tracking, and overall operational management within the Foodie online food ordering system.

CHAPTER 5. DISCUSSION

5.1. Project Outcomes

The developed system, FastFoodHub, is a web-based fast food ordering platform designed to simplify the food ordering process and improve the overall experience for both customers and restaurant administrators. The system successfully integrates essential features including food menu browsing, product searching, shopping cart management, online order placement, payment processing, invoice generation, and a comprehensive administrative management module.

Customers can browse food categories, view detailed product information, manage their shopping cart, and complete secure online transactions using either credit card or cash on delivery payment methods. The system also provides customers with the ability to manage their personal profiles, track their purchase history, and download or print order invoices for their records.

For restaurant administrators, FastFoodHub provides a centralized management dashboard that enables efficient control over food categories, products, customer orders, user accounts, customer inquiries, and sales reporting. The Admin Dashboard consolidates key performance indicators and operational tools into a single interface, improving administrative efficiency and supporting informed decision-making.

5.2. Performance Evaluation

Several aspects of the FastFoodHub system were evaluated to assess its usability, functionality, and reliability throughout the development process.

System Responsiveness

The system is implemented using ASP.NET Web Forms and Bootstrap, providing a responsive and consistent user interface across different screen sizes and devices. Core customer operations such as product browsing, cart management, and order placement were tested and performed reliably throughout the development phase. The administrative interface also demonstrated stable responsiveness during category management, order processing, and report generation tasks.

Data Reliability

The database is structured using a normalized relational design with MySQL, ensuring consistent data storage and accurate retrieval across all system modules. Relationships between entities such as users, products, categories, and orders are designed to maintain referential integrity and prevent data inconsistency. Product information, order records, and user account data are stored and retrieved accurately throughout the system.

Feature Integration

The system architecture supports seamless interaction between its core components. Customer-facing features such as the shopping cart, payment processing, and order tracking are integrated with the administrative back end, allowing administrators to monitor and manage orders in real time. The sales reporting module is connected to the order management system, enabling accurate generation of financial reports based on actual transaction data.

5.3. Discussion

5.3.1. Customer Features

The customer-facing module of FastFoodHub provides a streamlined and intuitive ordering experience. Users can register and log in to their accounts, browse food menus by category, view detailed product descriptions and pricing, add items to their shopping cart, and complete orders through a structured checkout process. Testing confirms that the shopping cart correctly calculates item totals and the grand total, while the checkout flow supports both credit card and cash on delivery payment methods without error.

The Purchase History and Invoice features further enhance the customer experience by providing transparent access to past transactions. Customers can view order details, monitor delivery statuses, and download or print PDF invoices for their records. These features improve trust and satisfaction by ensuring customers have complete visibility into their purchasing activity.

5.3.2. Administrative Management

The administrative module provides restaurant managers with a comprehensive toolset for managing all aspects of the system. The Admin Dashboard presents key business metrics briefly, reducing the time required for daily operational monitoring. Category and product management interfaces allow administrators to maintain an accurate and up-to-date menu, while the order status management module enables real-time processing and fulfillment tracking.

The user management and contact management interfaces further support administrators in maintaining system integrity and responding to customer inquiries effectively. Testing confirms that all administrative functions, including data creation, modification, deletion, and export operations, perform correctly and produce consistent results.

5.3.3. Sales Reporting

The Sales Report feature allows administrators to generate detailed financial reports by specifying a custom date range. The system accurately retrieves and displays transaction data including order details, customer information, product quantities, and revenue figures.

The Total SellingAmount indicator provides a quick overview of revenue for the selected period, while export options in CSV, PDF, and print formats support a variety of reporting and documentation needs.

CHAPTER 6. COMCLUSION AND FUTURE IMPROVEMENT

FastFoodHub has been successfully developed as a web-based fast food ordering system that addresses key challenges faced by both customers and restaurant administrators in managing online food ordering services. The system provides a complete set of features for the customer ordering workflow, from menu browsing and cart management to payment processing and invoice generation, as well as a robust administrative module for managing products, orders, users, and sales data.

Despite these achievements, several limitations remain in the current stage of development. First, the system is implemented as a desktop web application and does not yet include a dedicated mobile application, which may limit accessibility for users who primarily rely on smartphones. Second, the current payment integration uses a simulated credit card form and does not yet connect to a real payment gateway, which would be necessary for a production deployment. Third, the system does not yet include real-time notifications to inform customers about order status changes without requiring them to manually refresh the page.

In future development stages, several improvements can be implemented to enhance the system and address its current limitations:

- + Develop a mobile application version for iOS and Android platforms to improve accessibility and reach a wider user base.
- + Integrate a real payment gateway such as PayPal, Stripe, or a local payment provider to enable secure and authentic online transactions.
- + Implement real-time order status notifications using technologies such as SignalR or WebSockets to keep customers informed about their order progress without requiring page refreshes.
- + Add a product review and rating system to allow customers to provide feedback on food items, helping future customers make more informed choices and enabling administrators to identify popular or underperforming products.
- + Enhance the sales reporting module with visual analytics such as charts and graphs to present revenue trends, top-selling products, and customer behavior insights in a more accessible and actionable format.

Overall, FastFoodHub demonstrates how a web-based platform built with ASP.NET Web Forms, C#, and MySQL can effectively digitize and streamline the food ordering and restaurant management process. The system lays a solid foundation for future enhancements

and serves as a practical solution for small and medium-sized restaurants seeking to improve operational efficiency and customer satisfaction through digital transformation.

REFERENCES

- [1] [1] Microsoft, *ASP.NET Web Forms Overview*, Available: <https://learn.microsoft.com/en-us/aspnet/web-forms/>
- [2] [2] Microsoft, *C# Documentation*, Available: <https://learn.microsoft.com/en-us/dotnet/csharp/>
- [3] [3] Oracle Corporation, *MySQL Documentation*, Available: <https://dev.mysql.com/doc/>
- [4] [4] Bootstrap Team, *Bootstrap Documentation*, Available: <https://getbootstrap.com/docs/>
- [5] [5] Mozilla Developer Network (MDN), *JavaScript Guide*, Available: <https://developer.mozilla.org/en-US/docs/Web/JavaScript/Guide>
- [6] [6] Mozilla Developer Network (MDN), *HTML: HyperText Markup Language*, Available: <https://developer.mozilla.org/en-US/docs/Web/HTML>
- [7] [7] Mozilla Developer Network (MDN), *CSS: Cascading Style Sheets*, Available: <https://developer.mozilla.org/en-US/docs/Web/CSS>
- [8] [8] Microsoft, *Visual Studio IDE Documentation*, Available: <https://learn.microsoft.com/en-us/visualstudio/>
- [9] [9] R. Pressman and B. Maxim, *Software Engineering: A Practitioner's Approach*, 9th ed., McGraw-Hill Education, 2019.
- [10] [10] I. Sommerville, *Software Engineering*, 10th ed., Pearson Education, 2015.
- [11] [11] McDonald's, *Online Ordering System*, Available: <https://www.mcdonalds.com/>
- [12] [12] KFC, *Online Food Ordering System*, Available: <https://www.kfc.com/>
- [13] [13] Domino's Pizza, *Online Ordering Platform*, Available: <https://www.dominos.com/>
- [14] [14] Pizza Hut, *Online Ordering System*, Available: <https://www.pizzahut.com/>
- [15] [15] Burger King, *Online Food Ordering*, Available: <https://www.bk.com/>